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Mathematical Review

This book assumes some familiarity with elementary trigonometry, complex numbers, calculus, matrix algebra, and probability. Introductions to the first three topics by Chiang (1974) or Thomas (1972) are adequate; Marsden (1974) treated these issues in more depth. No matrix algebra is required beyond the level of standard econometrics texts such as Theil (1971) or Johnston (1984); for more detailed treatments, see O'Nan (1976), Strang (1976), and Magnus and Neudecker (1988). The concepts of probability and statistics from standard econometrics texts are also sufficient for getting through this book; for more complete introductions, see Lindgren (1976) or Hoel, Port, and Stone (1971).

This appendix reviews the necessary mathematical concepts and results. The reader familiar with these topics is invited to skip this material, or consult subheadings for desired coverage.

A.1. Trigonometry

Definitions

Figure A.1 displays a circle with unit radius centered at the origin in (x, y) -space. Let (x_0, y_0) denote some point on this unit circle, and consider the angle θ between this point and the x -axis. The *sine* of θ is defined as the y -coordinate of the point, and the *cosine* is the x -coordinate:

$$\sin(\theta) = y_0 \quad [\text{A.1.1}]$$

$$\cos(\theta) = x_0. \quad [\text{A.1.2}]$$

This text always measures angles in *radians*. The radian measure of the angle θ is defined as the distance traveled counterclockwise along the unit circle starting at the x -axis before reaching (x_0, y_0) . The circumference of a circle with unit radius is 2π . A rotation one-quarter of the way around the unit circle would therefore correspond to radian measure of $\theta = \frac{1}{4}(2\pi) = \pi/2$. An angle whose radian measure is $\pi/2$ is more commonly described as a right angle or a 90° angle. A 45° angle has radian measure of $\pi/4$, a 180° angle has radian measure of π , and so on.

Polar Coordinates

Consider a smaller triangle—say, the triangle with vertex (x_1, y_1) shown in Figure A.1—that shares the same angle θ as the original triangle with vertex

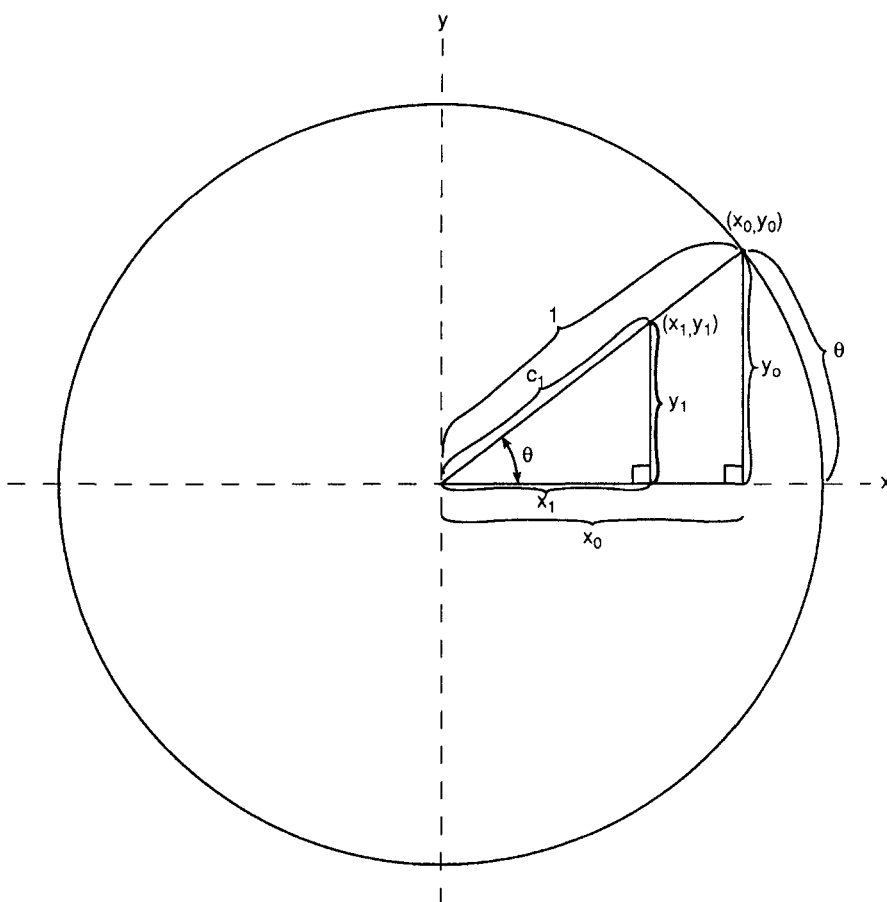


FIGURE A.1 Trigonometric functions as distances in (x, y) -space.

(x_0, y_0) . The ratio of any two sides of such a smaller triangle will be the same as that for the larger triangle:

$$y_1/c_1 = y_0/1 \quad [\text{A.1.3}]$$

$$x_1/c_1 = x_0/1. \quad [\text{A.1.4}]$$

Comparing [A.1.3] with [A.1.1], the y -coordinate of any point such as (x_1, y_1) in (x, y) -space may be expressed as

$$y_1 = c_1 \cdot \sin(\theta), \quad [\text{A.1.5}]$$

where c_1 is the distance from the origin to (x_1, y_1) and θ is the angle that the point (x_1, y_1) makes with the x -axis. Comparing [A.1.4] with [A.1.2], the x -coordinate of (x_1, y_1) can be expressed as

$$x_1 = c_1 \cdot \cos(\theta). \quad [\text{A.1.6}]$$

Recall further that the magnitude c_1 , which represents the distance from the origin to the point (x_1, y_1) , is given by the formula

$$c_1 = \sqrt{x_1^2 + y_1^2}. \quad [\text{A.1.7}]$$

Taking a point in (x, y) -space and writing it as $(c \cdot \cos(\theta), c \cdot \sin(\theta))$ is called describing the point in terms of its *polar coordinates* c and θ .

Properties of Sine and Cosine Functions

The functions $\sin(\theta)$ and $\cos(\theta)$ are called *trigonometric* or *sinusoidal* functions. Viewed as a function of θ , the sine function starts out at zero:

$$\sin(0) = 0.$$

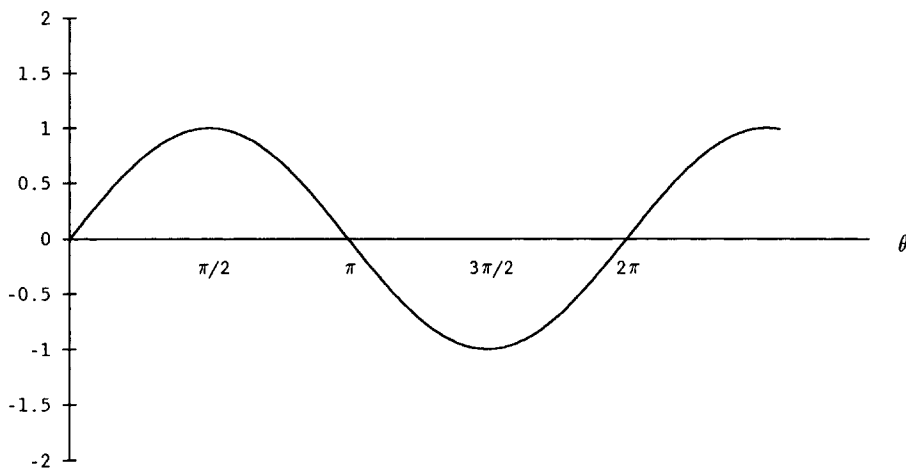
The sine function rises to 1 as θ increases to $\pi/2$ and then falls back to zero as θ increases further to π ; see panel (a) of Figure A.2. The function reaches its minimum value of -1 at $\theta = 3\pi/2$ and then begins climbing back up.

If we travel a distance of 2π radians around the unit circle, we are right back where we started, and the function repeats itself:

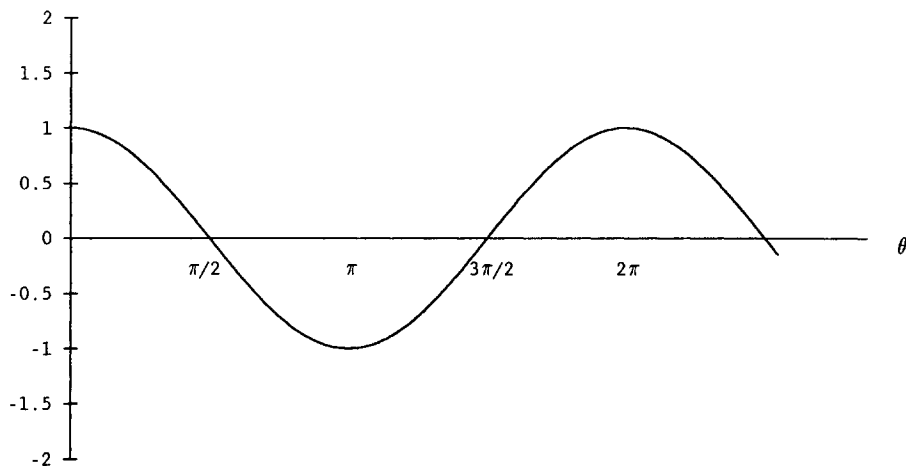
$$\sin(2\pi + \theta) = \sin(\theta).$$

The function would again repeat itself if we made two full revolutions around the unit circle. Indeed for any integer j ,

$$\sin(2\pi j + \theta) = \sin(\theta). \quad [\text{A.1.8}]$$



(a) $\sin(\theta)$



(b) $\cos(\theta)$

FIGURE A.2 Sine and cosine functions.

The sine function is thus *periodic* and is for this reason often useful for describing a time series that repeats itself in a particular cycle.

The cosine function starts out at unity and falls to zero as θ increases to $\pi/2$; see panel (b) of Figure A.2. It turns out simply to be a horizontal shift of the sine function:

$$\cos(\theta) = \sin\left(\theta + \frac{\pi}{2}\right). \quad [\text{A.1.9}]$$

The sine or cosine function can also be evaluated for negative values of θ , defined as a clockwise rotation around the unit circle from the x -axis. Clearly,

$$\sin(-\theta) = -\sin(\theta) \quad [\text{A.1.10}]$$

$$\cos(-\theta) = \cos(\theta). \quad [\text{A.1.11}]$$

For (x_0, y_0) a point on the unit circle, [A.1.7] implies that

$$1 = \sqrt{x_0^2 + y_0^2},$$

or, squaring both sides and using [A.1.1] and [A.1.2],

$$1 = [\cos(\theta)]^2 + [\sin(\theta)]^2. \quad [\text{A.1.12}]$$

Using Trigonometric Functions to Represent Cycles

Suppose we construct the function $g(\theta)$ by first multiplying θ by 2 and then evaluating the sine of the product:

$$g(\theta) = \sin(2\theta).$$

This doubles the frequency at which the function cycles. When θ goes from 0 to π , 2θ goes from 0 to 2π , and so $g(\theta)$ is back to its original value (see Figure A.3). In general, the function $\sin(k\theta)$ would go through k cycles in the time it takes $\sin(\theta)$ to complete a single cycle.

We will sometimes describe the value a variable y takes on at date t as a function of sines or cosines, such as

$$y_t = R \cdot \cos(\omega t + \alpha). \quad [\text{A.1.13}]$$

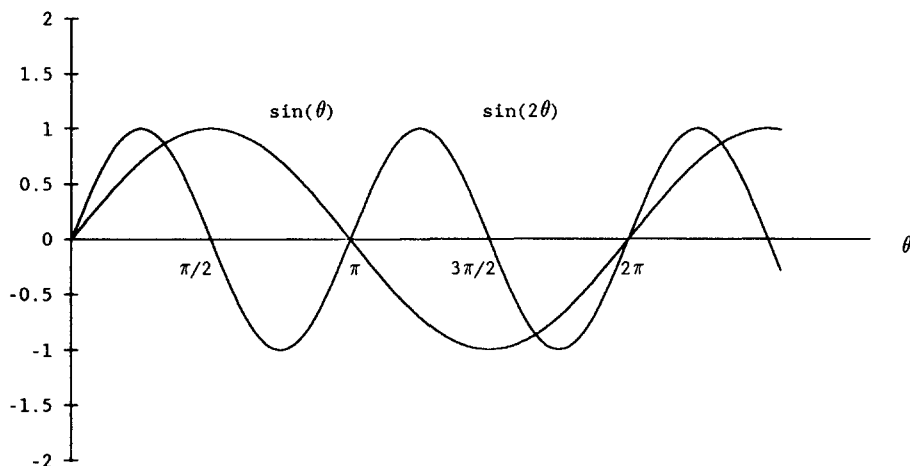


FIGURE A.3 Effect of changing frequency of a periodic function.

The parameter R gives the *amplitude* of [A.1.13]. The variable y_t will attain a maximum value of $+R$ and a minimum value of $-R$. The parameter α is the *phase*. The phase determines where in the cycle y_t would be at $t = 0$. The parameter ω governs how quickly the variable cycles, which can be summarized by either of two measures. The *period* is the length of time required for the process to repeat a full cycle. The period of [A.1.13] is $2\pi/\omega$. For example, if $\omega = 1$ then y repeats itself every 2π periods, whereas if $\omega = 2$ the process repeats itself every π periods. The *frequency* summarizes how frequently the process cycles compared with the simple function $\cos(t)$; thus, it measures the number of cycles completed during 2π periods. The frequency of $\cos(t)$ is unity, and the frequency of [A.1.13] is ω . For example, if $\omega = 2$, the cycles are completed twice as quickly as those for $\cos(t)$. There is a simple relation between these two measures of the speed of cycles—the period is equal to 2π divided by the frequency.

A.2. Complex Numbers

Definitions

Consider the following expression:

$$x^2 = 1. \quad [\text{A.2.1}]$$

There are two values of x that satisfy [A.2.1], namely, $x = 1$ and $x = -1$.

Suppose instead that we were given the following equation:

$$x^2 = -1. \quad [\text{A.2.2}]$$

No real number satisfies [A.2.2]. However, let us consider an imaginary number (denoted i) that does:

$$i^2 = -1. \quad [\text{A.2.3}]$$

We assume that i can be multiplied by a real number and manipulated using standard rules of algebra. For example,

$$2i + 3i = 5i$$

and

$$(2i) \cdot (3i) = (6)i^2 = -6.$$

This last property implies that a second solution to [A.2.2] is given by $x = -i$:

$$(-i)^2 = (-1)^2(i)^2 = -1.$$

Thus, [A.2.1] has two real roots ($+1$ and -1), whereas [A.2.2] has two imaginary roots (i and $-i$).

For any real numbers a and b , we can construct the expression

$$a + bi. \quad [\text{A.2.4}]$$

If $b = 0$, then [A.2.4] is a *real* number; whereas if $a = 0$ and b is nonzero, then [A.2.4] is an *imaginary* number. A number written in the general form of [A.2.4] is called a *complex* number.

Rules for Manipulating Complex Numbers

Complex numbers are manipulated using standard rules of algebra. Two complex numbers are added as follows:

$$(a_1 + b_1i) + (a_2 + b_2i) = (a_1 + a_2) + (b_1 + b_2)i.$$

Complex numbers are multiplied this way:

$$\begin{aligned}(a_1 + b_1i) \cdot (a_2 + b_2i) &= a_1a_2 + a_1b_2i + b_1a_2i + b_1b_2i^2 \\ &= (a_1a_2 - b_1b_2) + (a_1b_2 + b_1a_2)i.\end{aligned}$$

Note that the resulting expressions are always simplified by separating the real component (such as $[a_1a_2 - b_1b_2]$) from the imaginary component (such as $[a_1b_2 + b_1a_2]i$).

Graphical Representation of Complex Numbers

A complex number $(a + bi)$ is sometimes represented graphically in an *Argand diagram* as in Figure A.4. The value of the real component (a) is plotted on the horizontal axis, and the imaginary component (b) is plotted on the vertical axis. The size, or *modulus*, of a complex number is measured the same way as the distance from the origin of a real element in (x, y) -space (see equation [A.1.7]):

$$|a + bi| \equiv \sqrt{a^2 + b^2}. \quad [\text{A.2.5}]$$

The *complex unit circle* is the set of all complex numbers whose modulus is 1. For example, the real number $+1$ is on the complex unit circle (represented by

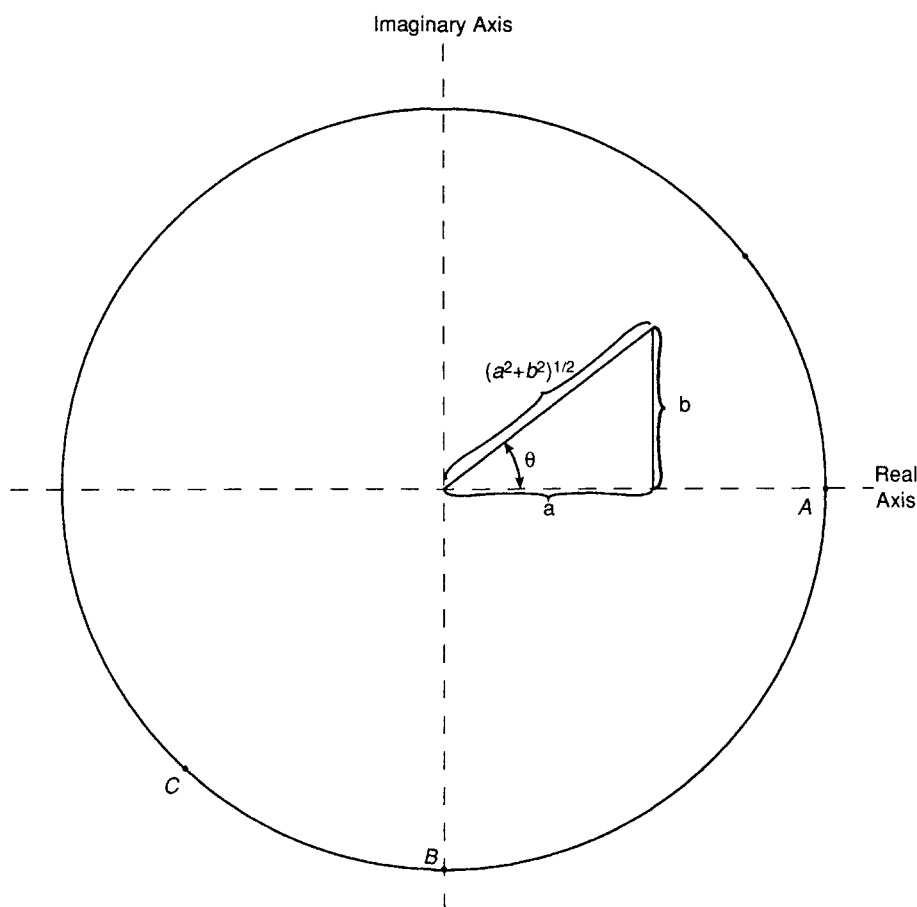


FIGURE A.4 Argand diagram and the complex unit circle.

the point A in Figure A.4). So are the imaginary number $-i$ (point B) and the complex number $(-0.6 - 0.8i)$ (point C).

We will often be interested in whether a complex number is less than 1 in modulus, in which case the number is said to be inside the unit circle. For example, $(-0.3 + 0.4i)$ has modulus 0.5, so it lies inside the unit circle, whereas $(3 + 4i)$, with modulus 5, lies outside the unit circle.

Polar Coordinates

Just as a point in (x, y) -space can be represented by its distance c from the origin and its angle θ with the x -axis, the complex number $a + bi$ can be represented by the distance of (a, b) from the origin (the modulus of the complex number),

$$R = \sqrt{a^2 + b^2},$$

and by the angle θ that the point (a, b) makes with the real axis, characterized by

$$\begin{aligned}\cos(\theta) &= a/R \\ \sin(\theta) &= b/R.\end{aligned}$$

Thus, the complex number $a + bi$ is written in polar coordinate form as

$$[R \cdot \cos(\theta) + i \cdot R \cdot \sin(\theta)] = R[\cos(\theta) + i \cdot \sin(\theta)]. \quad [\text{A.2.6}]$$

Complex Conjugates

The *complex conjugate* of $(a + bi)$ is given by $(a - bi)$. The numbers $(a + bi)$ and $(a - bi)$ are described as a *conjugate pair*. Notice that adding a conjugate pair produces a real result:

$$(a + bi) + (a - bi) = 2a.$$

The product of a conjugate pair is also real:

$$(a + bi) \cdot (a - bi) = a^2 + b^2. \quad [\text{A.2.7}]$$

Comparing this with [A.2.5], we see that the modulus of a complex number $(a + bi)$ can be thought of as the square root of the product of the number with its complex conjugate:

$$|a + bi| = \sqrt{(a + bi)(a - bi)}. \quad [\text{A.2.8}]$$

Quadratic Equations

A *quadratic equation*

$$\alpha x^2 + \beta x + \gamma = 0 \quad [\text{A.2.9}]$$

with $\alpha \neq 0$ has two solutions:

$$x_1 = \frac{-\beta + (\beta^2 - 4\alpha\gamma)^{1/2}}{2\alpha} \quad [\text{A.2.10}]$$

$$x_2 = \frac{-\beta - (\beta^2 - 4\alpha\gamma)^{1/2}}{2\alpha}. \quad [\text{A.2.11}]$$

When $(\beta^2 - 4\alpha\gamma) \geq 0$, both these roots are real, whereas when $(\beta^2 - 4\alpha\gamma) < 0$, the roots are complex. Notice that when the roots are complex they appear as a

conjugate pair:

$$\begin{aligned}x_1 &= \{-\beta/[2\alpha]\} + \{(1/[2\alpha])(4\alpha\gamma - \beta^2)^{1/2}\}i \\x_2 &= \{-\beta/[2\alpha]\} - \{(1/[2\alpha])(4\alpha\gamma - \beta^2)^{1/2}\}i.\end{aligned}$$

A.3. Calculus

Continuity

A function $f(x)$ is said to be *continuous* at $x = c$ if $f(c)$ is finite and if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ whenever $|x - c| < \delta$.

Derivatives of Some Simple Functions

The *derivative* of $f(\cdot)$ with respect to x is defined by

$$\frac{df}{dx} = \lim_{\Delta \rightarrow 0} \frac{f(x + \Delta) - f(x)}{\Delta},$$

provided that this limit exists.

If $f(\cdot)$ is linear in x , or

$$f(x) = \alpha + \beta x,$$

then the derivative is just the coefficient on x :

$$\frac{df}{dx} = \lim_{\Delta \rightarrow 0} \frac{[\alpha + \beta(x + \Delta)] - [\alpha + \beta x]}{\Delta} = \lim_{\Delta \rightarrow 0} \frac{\beta\Delta}{\Delta} = \beta.$$

For a quadratic function

$$f(x) = x^2,$$

the derivative is

$$\begin{aligned}\frac{df}{dx} &= \lim_{\Delta \rightarrow 0} \frac{[x + \Delta]^2 - x^2}{\Delta} \\&= \lim_{\Delta \rightarrow 0} \frac{[x^2 + 2x\Delta + \Delta^2] - x^2}{\Delta} \\&= \lim_{\Delta \rightarrow 0} \{2x + \Delta\} \\&= 2x,\end{aligned}$$

and in general,

$$\frac{dx^k}{dx} = kx^{k-1}. \quad [\text{A.3.1}]$$

For the trigonometric functions, it can be shown that when x is measured in radians,

$$\frac{d \sin(x)}{dx} = \cos(x) \quad [\text{A.3.2}]$$

$$\frac{d \cos(x)}{dx} = -\sin(x). \quad [\text{A.3.3}]$$

The derivative $df(x)/dx$ is itself a function of x . Often we want to specify the point at which the derivative should be evaluated, say, c . This is indicated by

$$\left. \frac{df(x)}{dx} \right|_{x=c}.$$

For example,

$$\left. \frac{dx^2}{dx} \right|_{x=3} = 2x|_{x=3} = 6.$$

Note that this notation refers to taking the derivative first and then evaluating the derivative at a particular point such as $x = 3$.

Chain Rule

The *chain rule* states that for composite functions such as

$$g(x) = f(u(x)),$$

the derivative is

$$\frac{dg(x)}{dx} = \frac{df}{du} \cdot \frac{du}{dx}. \quad [\text{A.3.4}]$$

For example, to evaluate

$$\frac{d(\alpha + \beta x)^k}{dx},$$

we let $f(u) = u^k$ and $u(x) = \alpha + \beta x$. Then

$$\frac{df}{du} \cdot \frac{du}{dx} = ku^{k-1} \cdot \beta.$$

Thus,

$$\frac{d(\alpha + \beta x)^k}{dx} = \beta k(\alpha + \beta x)^{k-1}.$$

Higher-Order Derivatives

The *second derivative* is defined by

$$\frac{d^2f(x)}{dx^2} = \frac{d}{dx} \left[\frac{df(x)}{dx} \right].$$

For example,

$$\frac{d^2x^k}{dx^2} = \frac{d[kx^{k-1}]}{dx} = k(k-1)x^{k-2}$$

and

$$\frac{d^2 \sin(x)}{dx^2} = \frac{d \cos(x)}{dx} = -\sin(x). \quad [\text{A.3.5}]$$

In general, the j th-order derivative is the derivative of the $(j-1)$ th-order derivative.

Geometric Series

Consider the sum

$$s_T = 1 + \phi + \phi^2 + \phi^3 + \cdots + \phi^T. \quad [\text{A.3.6}]$$

Multiplying both sides of [A.3.6] by ϕ ,

$$\phi s_T = \phi + \phi^2 + \phi^3 + \cdots + \phi^T + \phi^{T+1}. \quad [\text{A.3.7}]$$

Subtracting [A.3.7] from [A.3.6] produces

$$(1 - \phi)s_T = 1 - \phi^{T+1}. \quad [\text{A.3.8}]$$

For any $\phi \neq 1$, both sides of [A.3.8] can be divided by $(1 - \phi)$. Hence, the sum in [A.3.6] is equal to

$$s_T = \begin{cases} \frac{1 - \phi^{T+1}}{1 - \phi} & \phi \neq 1 \\ T + 1 & \phi = 1. \end{cases} \quad [\text{A.3.9}]$$

From [A.3.9],

$$\lim_{T \rightarrow \infty} s_T = \frac{1}{1 - \phi} \quad |\phi| < 1,$$

and so

$$(1 + \phi + \phi^2 + \phi^3 + \cdots) = \frac{1}{1 - \phi} \quad |\phi| < 1. \quad [\text{A.3.10}]$$

Taylor Series Approximations

Suppose that the first through the $(r + 1)$ th derivatives of a function $f(x)$ exist and are continuous in a neighborhood of c . *Taylor's theorem* states that the value of $f(x)$ at $x = c + \Delta$ is given by

$$\begin{aligned} f(c + \Delta) = f(c) &+ \left. \frac{df}{dx} \right|_{x=c} \cdot \Delta + \frac{1}{2!} \left. \frac{d^2f}{dx^2} \right|_{x=c} \cdot \Delta^2 \\ &+ \frac{1}{3!} \left. \frac{d^3f}{dx^3} \right|_{x=c} \cdot \Delta^3 + \cdots + \frac{1}{r!} \left. \frac{d^rf}{dx^r} \right|_{x=c} \cdot \Delta^r + R_r(c, x), \end{aligned} \quad [\text{A.3.11}]$$

where $r!$ denotes r factorial:

$$r! \equiv r(r - 1) \cdot (r - 2) \cdots 2 \cdot 1.$$

The remainder $R_r(c, x)$ is given by

$$R_r(c, x) = \frac{1}{(r + 1)!} \cdot \left. \frac{d^{r+1}f}{dx^{r+1}} \right|_{x=\delta} \cdot \Delta^{r+1},$$

where δ is a number between c and x . Notice that the remainder vanishes for small Δ :

$$\lim_{\Delta \rightarrow 0} \frac{R_r(c, x)}{\Delta^r} = 0.$$

Setting $R_r(c, x) = 0$ and $x = c + \Delta$ in [A.3.11] produces an *rth-order Taylor series approximation* to the function $f(x)$ in the neighborhood of $x = c$:

$$f(x) \cong f(c) + \left. \frac{df}{dx} \right|_{x=c} \cdot (x - c) + \frac{1}{2!} \left. \frac{d^2f}{dx^2} \right|_{x=c} \cdot (x - c)^2 + \cdots + \frac{1}{r!} \left. \frac{d^rf}{dx^r} \right|_{x=c} \cdot (x - c)^r. \quad [\text{A.3.12}]$$

Power Series

If the remainder $R_r(c, x)$ in [A.3.11] converges to zero for all x as $r \rightarrow \infty$, a power series can be used to characterize the function $f(x)$. To find a power series, we choose a particular value c around which to center the expansion, such as $c = 0$. We then use [A.3.12] with $r \rightarrow \infty$. For example, consider the sine function. The first two derivatives are given by [A.3.2] and [A.3.5], with the following higher-order derivatives:

$$\frac{d^3 \sin(x)}{dx^3} = -\cos(x)$$

$$\frac{d^4 \sin(x)}{dx^4} = \sin(x)$$

$$\frac{d^5 \sin(x)}{dx^5} = \cos(x),$$

and so on. Evaluated at $x = 0$, we have

$$f(0) = \sin(0) = 0$$

$$\left. \frac{df}{dx} \right|_{x=0} = \cos(0) = 1$$

$$\left. \frac{d^2f}{dx^2} \right|_{x=0} = -\sin(0) = 0$$

$$\left. \frac{d^3f}{dx^3} \right|_{x=0} = -\cos(0) = -1$$

$$\left. \frac{d^4f}{dx^4} \right|_{x=0} = \sin(0) = 0$$

$$\left. \frac{d^5f}{dx^5} \right|_{x=0} = \cos(0) = 1.$$

Substituting into [A.3.12] with $c = 0$ and letting $r \rightarrow \infty$ produces a power series for the sine function:

$$\sin(x) = x - \frac{1}{3!} x^3 + \frac{1}{5!} x^5 - \frac{1}{7!} x^7 + \cdots. \quad [\text{A.3.13}]$$

Similar calculations give a power series for the cosine function:

$$\cos(x) = 1 - \frac{1}{2!} x^2 + \frac{1}{4!} x^4 - \frac{1}{6!} x^6 + \cdots. \quad [\text{A.3.14}]$$

Exponential Functions

A number γ raised to the power x ,

$$f(x) = \gamma^x,$$

is called an *exponential function* of x . The number γ is called the *base* of this function, and x is called the *exponent*. To multiply two exponential functions that share the same base, the exponents are added:

$$(\gamma^x) \cdot (\gamma^y) = \gamma^{(x+y)}. \quad [\text{A.3.15}]$$

For example,

$$(\gamma^2) \cdot (\gamma^3) = (\gamma \cdot \gamma) \cdot (\gamma \cdot \gamma \cdot \gamma) = \gamma^5.$$

To raise an exponential function to the power k , the exponents are multiplied:

$$[\gamma^x]^k = \gamma^{xk}. \quad [\text{A.3.16}]$$

For example,

$$[\gamma^2]^3 = [\gamma^2] \cdot [\gamma^2] \cdot [\gamma^2] = \gamma^6.$$

Exponentiation is distributive over multiplication:

$$(\alpha \cdot \beta)^x = (\alpha^x) \cdot (\beta^x). \quad [\text{A.3.17}]$$

Negative exponents denote reciprocals:

$$\gamma^{-k} = (1/\gamma^k).$$

Any number raised to the power 0 is taken to be equal to unity:

$$\gamma^0 = 1. \quad [\text{A.3.18}]$$

This convention is sensible, since if $y = -x$ in [A.3.15],

$$(\gamma^x)(\gamma^{-x}) = \gamma^0$$

and

$$(\gamma^x)(\gamma^{-x}) = \frac{\gamma^x}{\gamma^x} = 1.$$

The Number e

The base for the natural logarithms is denoted e . The number e has the property that an exponential function with base e equals its own derivative:

$$\frac{de^x}{dx} = e^x. \quad [\text{A.3.19}]$$

Clearly, all the higher-order derivatives of e^x are equal to e^x as well:

$$\frac{d^r e^x}{dx^r} = e^x. \quad [\text{A.3.20}]$$

We sometimes use the expression “exp[x]” to represent “ e raised to the power x ”:

$$\exp[x] \equiv e^x.$$

If $u(x)$ denotes a separate function of x , the derivative of the compound function $e^{u(x)}$ can be evaluated using the chain rule:

$$\frac{de^{u(x)}}{dx} = \frac{de^u}{du} \cdot \frac{du}{dx} = e^{u(x)} \frac{du}{dx}. \quad [\text{A.3.21}]$$

To find a power series for the function $f(x) = e^x$, notice from [A.3.20] that

$$\frac{d^r f}{dx^r} = e^x,$$

and so, from [A.3.18],

$$\left. \frac{d^r f}{dx^r} \right|_{x=0} = e^0 = 1 \quad [\text{A.3.22}]$$

for all r . Substituting [A.3.22] into [A.3.12] with $c = 0$ yields a power series for the function $f(x) = e^x$:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots. \quad [\text{A.3.23}]$$

Setting $x = 1$ in [A.3.23] gives a numerical procedure for calculating the value of e :

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots = 2.71828 \dots$$

Euler Relations and De Moivre's Theorem

Suppose we evaluate the power series [A.3.23] at the imaginary number $x = i\theta$, where $i = \sqrt{-1}$ and θ is some real angle measured in radians:

$$\begin{aligned} e^{i\theta} &= 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \cdots \\ &= \left\{ 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots \right\} + i \cdot \left\{ \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots \right\}. \end{aligned} \quad [\text{A.3.24}]$$

Reflecting on [A.3.13] and [A.3.14] gives another interpretation of [A.3.24]:

$$e^{i\theta} = \cos(\theta) + i \cdot \sin(\theta). \quad [\text{A.3.25}]$$

Similarly,

$$\begin{aligned} e^{-i\theta} &= 1 + (-i\theta) + \frac{(-i\theta)^2}{2!} + \frac{(-i\theta)^3}{3!} + \frac{(-i\theta)^4}{4!} + \frac{(-i\theta)^5}{5!} + \cdots \\ &= \left\{ 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots \right\} - i \cdot \left\{ \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots \right\} \\ &= \cos(\theta) - i \cdot \sin(\theta). \end{aligned} \quad [\text{A.3.26}]$$

To raise a complex number $(a + bi)$ to the k th power, the complex number is written in polar coordinate form as in [A.2.6]:

$$a + bi = R[\cos(\theta) + i \cdot \sin(\theta)].$$

Using [A.3.25], this can then be treated as an exponential function of θ :

$$a + bi = R \cdot e^{i\theta}. \quad [\text{A.3.27}]$$

Now raise both sides of [A.3.27] to the k th power, recalling [A.3.17] and [A.3.16]:

$$(a + bi)^k = R^k \cdot [e^{i\theta}]^k = R^k \cdot e^{i\theta k}. \quad [\text{A.3.28}]$$

Finally, use [A.3.25] in reverse,

$$e^{i(\theta k)} = \cos(\theta k) + i \cdot \sin(\theta k),$$

to deduce that [A.3.28] can be written

$$(a + bi)^k = R^k \cdot [\cos(\theta k) + i \cdot \sin(\theta k)]. \quad [\text{A.3.29}]$$

Definition of Natural Logarithm

The *natural logarithm* (denoted throughout the text simply by “log”) is the inverse of the function e^x :

$$\log(e^x) \equiv x.$$

Notice from [A.3.18] that $e^0 = 1$ and therefore $\log(1) = 0$.

Properties of Logarithms

For any $x > 0$, it is also the case that

$$x = e^{\log(x)}. \quad [\text{A.3.30}]$$

From [A.3.30] and [A.3.15], we see that the log of the product of two numbers is equal to the sum of the logs:

$$\log(a \cdot b) = \log[(e^{\log(a)}) \cdot (e^{\log(b)})] = \log[e^{(\log(a) + \log(b))}] = \log(a) + \log(b).$$

Also, use [A.3.16] to write

$$x^a = [e^{\log(x)}]^a = e^{a \cdot \log(x)}. \quad [\text{A.3.31}]$$

Taking logs of both sides of [A.3.31] reveals that the log of a number raised to the a power is equal to a times the log of the number:

$$\log(x^a) = a \cdot \log(x).$$

Derivatives of Natural Logarithms

Let $u(x) = \log(x)$, and write the right side of [A.3.30] as $e^{u(x)}$. Differentiating both sides of [A.3.30] using [A.3.21] reveals that

$$\frac{dx}{dx} = e^{\log(x)} \cdot \frac{d \log(x)}{dx}$$

or

$$1 = x \frac{d \log(x)}{dx}.$$

Thus,

$$\frac{d \log(x)}{dx} = \frac{1}{x}. \quad [\text{A.3.32}]$$

Logarithms and Elasticities

It is sometimes also useful to differentiate a function $f(x)$ with respect to the variable $\log(x)$. To do so, write $f(x)$ as $f(u(x))$, where

$$u(x) = \exp[\log(x)].$$

Now use the chain rule to differentiate:

$$\frac{df(x)}{d \log(x)} = \frac{df}{du} \cdot \frac{du}{d \log(x)}. \quad [\text{A.3.33}]$$

But from [A.3.21],

$$\frac{du}{d \log(x)} = \exp[\log(x)] \frac{d \log(x)}{d \log(x)} = x. \quad [\text{A.3.34}]$$

Substituting [A.3.34] into [A.3.33] gives

$$\frac{df(x)}{d \log(x)} = x \frac{df}{dx}.$$

It follows from [A.3.32] that

$$\frac{d \log f(x)}{d \log(x)} = \frac{1}{f} x \frac{df}{dx} \cong \frac{[f(x + \Delta) - f(x)]/f(x)}{[(x + \Delta) - x]/x},$$

which has the interpretation as the *elasticity* of f with respect to x , or the percent change in f resulting from a 1% increase in x .

Logarithms and Percent

An approximation to the natural log function is obtained from a first-order Taylor series around $c = 1$:

$$\log(1 + \Delta) \cong \log(1) + \left. \frac{d \log(x)}{dx} \right|_{x=1} \cdot \Delta. \quad [\text{A.3.35}]$$

But $\log(1) = 0$, and

$$\left. \frac{d \log(x)}{dx} \right|_{x=1} = \left. \frac{1}{x} \right|_{x=1} = 1.$$

Thus, for Δ close to zero, an excellent approximation is provided by

$$\log(1 + \Delta) \cong \Delta. \quad [\text{A.3.36}]$$

An implication of [A.3.36] is the following. Let r denote the net interest rate measured as a fraction of 1; for example, $r = 0.05$ corresponds to a 5% interest rate. Then $(1 + r)$ denotes the gross interest rate (principal plus net interest). Equation [A.3.36] says that the log of the gross interest rate $(1 + r)$ is essentially the same number as the net interest rate (r).

Definition of Indefinite Integral

Integration (indicated by $\int dx$) is the inverse operation from differentiation. For example,

$$\int x \, dx = x^2/2, \quad [\text{A.3.37}]$$

because

$$\frac{d(x^2/2)}{dx} = x. \quad [\text{A.3.38}]$$

The function $x^2/2$ is not the only function satisfying [A.3.38]; the function

$$(x^2/2) + C$$

also works for any constant C . The term C is referred to as the *constant of integration*.

Some Useful Indefinite Integrals

The following integrals can be confirmed from [A.3.1], [A.3.32], [A.3.2], [A.3.3], and [A.3.21]:

$$\int x^k dx = \frac{x^{k+1}}{k+1} + C \quad k \neq -1 \quad [\text{A.3.39}]$$

$$\int x^{-1} dx = \begin{cases} \log(x) + C & x > 0 \\ \log(-x) + C & x < 0 \end{cases} \quad [\text{A.3.40}]$$

$$\int \cos(x) dx = \sin(x) + C \quad [\text{A.3.41}]$$

$$\int \sin(x) dx = -\cos(x) + C \quad [\text{A.3.42}]$$

$$\int e^{ax} dx = (1/a) \cdot e^{ax} + C. \quad [\text{A.3.43}]$$

It is also straightforward to demonstrate that for constants a and b not depending on x ,

$$\int [a \cdot f(x) + b \cdot g(x)] dx = a \int f(x) dx + b \int g(x) dx + C.$$

Definite Integrals

Consider the continuous function $f(x)$ plotted in Figure A.5. Define the function $A(x; a)$ to be the area under $f(x)$ between a and x , viewed as a function of x . Thus, $A(b; a)$ would be the area between a and b . Suppose we increase b by a small amount Δ . This is approximately the same as adding a rectangle of height $f(b)$ and width Δ to the area $A(b; a)$:

$$A(b + \Delta; a) \cong A(b; a) + f(b) \cdot \Delta,$$

or

$$\frac{A(b + \Delta; a) - A(b; a)}{\Delta} \cong f(b).$$

In the limit as $\Delta \rightarrow 0$,

$$\left. \frac{dA(x; a)}{dx} \right|_{x=b} = f(b). \quad [\text{A.3.44}]$$

Now, [A.3.44] has to hold for any value of $b > a$ that we might have chosen,

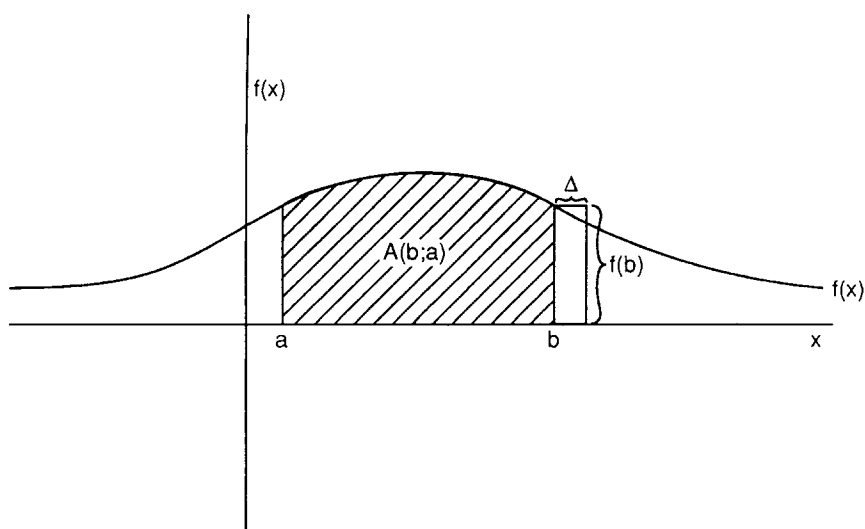


FIGURE A.5 The definite integral as the area under a function.

implying that the area function $A(x; a)$ is the inverse of differentiation:

$$A(x; a) = F(x) + C, \quad [\text{A.3.45}]$$

where

$$\frac{dF(x)}{dx} = f(x).$$

To find the value of C , notice that $A(a; a)$ in [A.3.45] should be equal to zero:

$$A(a; a) = 0 = F(a) + C.$$

For this to be true,

$$C = -F(a). \quad [\text{A.3.46}]$$

Evaluating [A.3.45] at $x = b$, the area between a and b is given by

$$A(b; a) = F(b) + C;$$

or using [A.3.46],

$$A(b; a) = F(b) - F(a), \quad [\text{A.3.47}]$$

where $F(x)$ satisfies $dF/dx = f(x)$:

$$F(x) = \int f(x) dx.$$

Equation [A.3.47] is known as the *fundamental theorem of calculus*.

The operation in [A.3.47] is known as calculating a *definite integral*:

$$\int_a^b f(x) dx \equiv \left[\int f(x) dx \right] \Big|_{x=b} - \left[\int f(x) dx \right] \Big|_{x=a}.$$

For example, to find the area under the sine function between $\theta = 0$ and $\theta = \pi/2$, we use [A.3.42]:

$$\begin{aligned}
 \int_0^{\pi/2} \sin(x) \, dx &= [-\cos(x)]|_{x=\pi/2} - [-\cos(x)]|_{x=0} \\
 &= [-\cos(\pi/2)] + [\cos(0)] \\
 &= 0 + 1 \\
 &= 1.
 \end{aligned}$$

To find the area between 0 and 2π , we take

$$\begin{aligned}
 \int_0^{2\pi} \sin(x) \, dx &= [-\cos(2\pi)] + \cos(0) \\
 &= -1 + 1 \\
 &= 0.
 \end{aligned}$$

The positive values for $\sin(x)$ between 0 and π exactly cancel out the negative values between π and 2π .

A.4. Matrix Algebra

Definitions

An $(m \times n)$ matrix is an array of numbers ordered into m rows and n columns:

$$\mathbf{A}_{(m \times n)} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

If there is only one column ($n = 1$), then $\mathbf{A}$ is described as a *column vector*, whereas with only one row ($m = 1$), $\mathbf{A}$ is called a *row vector*. A single number ($n = 1$ and $m = 1$) is called a *scalar*.

If the number of rows equals the number of columns ($m = n$), the matrix is said to be *square*. The diagonal running through $(a_{11}, a_{22}, \dots, a_{nn})$ in a square matrix is called the *principal diagonal*. If all elements off the principal diagonal are zero, the matrix is said to be *diagonal*.

A matrix is sometimes specified by describing the element in row i , column j :

$$\mathbf{A} = [a_{ij}].$$

Summation and Multiplication

Two $(m \times n)$ matrices are added element by element:

$$\begin{aligned}
 \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix} \\
 = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix};
 \end{aligned}$$

or, more compactly,

$$\underset{(m \times n)}{\mathbf{A}} + \underset{(m \times n)}{\mathbf{B}} = [a_{ij} + b_{ij}].$$

The product of an $(m \times n)$ matrix and an $(n \times q)$ matrix is an $(m \times q)$ matrix:

$$\underset{(m \times n)}{\mathbf{A}} \times \underset{(n \times q)}{\mathbf{B}} = \underset{(m \times q)}{\mathbf{C}},$$

where the row i , column j element of $\mathbf{C}$ is given by $\sum_{k=1}^n a_{ik} b_{kj}$. Notice that multiplication requires that the number of columns of $\mathbf{A}$ be the same as the number of rows of $\mathbf{B}$.

To multiply $\mathbf{A}$ by a scalar α , each element of $\mathbf{A}$ is multiplied by α :

$$\underset{(1 \times 1)}{\alpha} \times \underset{(m \times n)}{\mathbf{A}} = \underset{(m \times n)}{\mathbf{C}},$$

with

$$\mathbf{C} = [\alpha a_{ij}].$$

It is easy to show that addition is commutative:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A};$$

whereas multiplication is not:

$$\mathbf{AB} \neq \mathbf{BA}.$$

Indeed, the product $\mathbf{BA}$ will not exist unless $m = q$, and even where it exists, $\mathbf{AB}$ would be equal to $\mathbf{BA}$ only in rather special cases.

Both addition and multiplication are associative:

$$\begin{aligned} (\mathbf{A} + \mathbf{B}) + \mathbf{C} &= \mathbf{A} + (\mathbf{B} + \mathbf{C}) \\ (\mathbf{AB})\mathbf{C} &= \mathbf{A}(\mathbf{BC}). \end{aligned}$$

Identity Matrix

The *identity matrix* of order n (denoted $\mathbf{I}_n$) is an $(n \times n)$ matrix with 1s along the principal diagonal and 0s elsewhere:

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

For any $(m \times n)$ matrix $\mathbf{A}$,

$$\mathbf{A} \times \mathbf{I}_n = \mathbf{A}$$

and also

$$\mathbf{I}_m \times \mathbf{A} = \mathbf{A}.$$

Powers of Matrices

For an $(n \times n)$ matrix $\mathbf{A}$, the expression $\mathbf{A}^2$ denotes $\mathbf{A} \cdot \mathbf{A}$. The expression $\mathbf{A}^k$ indicates the matrix $\mathbf{A}$ multiplied by itself k times, with $\mathbf{A}^0$ interpreted as the $(n \times n)$ identity matrix.

Transposition

Let a_{ij} denote the row i , column j element of a matrix $\mathbf{A}$:

$$\mathbf{A} = [a_{ij}].$$

The *transpose* of $\mathbf{A}$ (denoted $\mathbf{A}'$) is given by

$$\mathbf{A}' = [a_{ji}].$$

For example, the transpose of

$$\begin{bmatrix} 2 & 4 & 6 \\ 3 & 5 & 7 \\ 1 & 2 & 3 \end{bmatrix}$$

is

$$\begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 2 \\ 6 & 7 & 3 \end{bmatrix}.$$

The transpose of a row vector is a column vector.

It is easy to verify the following:

$$(\mathbf{A}')' = \mathbf{A} \quad [\text{A.4.1}]$$

$$(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}' \quad [\text{A.4.2}]$$

$$(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'. \quad [\text{A.4.3}]$$

Symmetric Matrices

A square matrix satisfying $\mathbf{A} = \mathbf{A}'$ is said to be *symmetric*.

Trace of a Matrix

The *trace* of an $(n \times n)$ matrix is defined as the sum of the elements along the principal diagonal:

$$\text{trace}(\mathbf{A}) \equiv a_{11} + a_{22} + \cdots + a_{nn}.$$

If $\mathbf{A}$ is an $(m \times n)$ matrix and $\mathbf{B}$ is an $(n \times m)$ matrix, then $\mathbf{AB}$ is an $(m \times m)$ matrix whose trace is

$$\text{trace}(\mathbf{AB}) = \sum_{j=1}^n a_{1j}b_{j1} + \sum_{j=1}^n a_{2j}b_{j2} + \cdots + \sum_{j=1}^n a_{mj}b_{jm} = \sum_{k=1}^m \sum_{j=1}^n a_{kj}b_{jk}.$$

The product $\mathbf{BA}$ is an $(n \times n)$ matrix whose trace is

$$\text{trace}(\mathbf{BA}) = \sum_{k=1}^m b_{1k}a_{k1} + \sum_{k=1}^m b_{2k}a_{k2} + \cdots + \sum_{k=1}^m b_{nk}a_{kn} = \sum_{j=1}^n \sum_{k=1}^m b_{jk}a_{kj}.$$

Thus,

$$\text{trace}(\mathbf{AB}) = \text{trace}(\mathbf{BA}).$$

If $\mathbf{A}$ and $\mathbf{B}$ are both $(n \times n)$ matrices, then

$$\text{trace}(\mathbf{A} + \mathbf{B}) = \text{trace}(\mathbf{A}) + \text{trace}(\mathbf{B}).$$

If $\mathbf{A}$ is an $(n \times n)$ matrix and λ is a scalar, then

$$\text{trace}(\lambda \mathbf{A}) = \sum_{i=1}^n \lambda a_{ii} = \lambda \cdot \sum_{i=1}^n a_{ii} = \lambda \cdot \text{trace}(\mathbf{A}).$$

Partitioned Matrices

A *partitioned matrix* is a matrix whose individual elements are themselves matrices. For example, the (3×4) matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

could be written as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{a}'_1 & \mathbf{a}'_2 \end{bmatrix}$$

where

$$\begin{aligned} \mathbf{A}_1 &\equiv \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} & \mathbf{A}_2 &\equiv \begin{bmatrix} a_{13} & a_{14} \\ a_{23} & a_{24} \end{bmatrix} \\ \mathbf{a}'_1 &\equiv [a_{31} \quad a_{32}] & \mathbf{a}'_2 &\equiv [a_{33} \quad a_{34}]. \end{aligned}$$

Partitioned matrices are added or multiplied as if the individual elements were scalars, provided that the row and column dimensions permit the appropriate matrix operations. For example,

$$\begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{A}_3 & \mathbf{A}_4 \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 & \mathbf{B}_2 \\ \mathbf{B}_3 & \mathbf{B}_4 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 + \mathbf{B}_1 & \mathbf{A}_2 + \mathbf{B}_2 \\ \mathbf{A}_3 + \mathbf{B}_3 & \mathbf{A}_4 + \mathbf{B}_4 \end{bmatrix}.$$

Similarly,

$$\begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{A}_3 & \mathbf{A}_4 \end{bmatrix} \times \begin{bmatrix} \mathbf{B}_1 & \mathbf{B}_2 \\ \mathbf{B}_3 & \mathbf{B}_4 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1\mathbf{B}_1 + \mathbf{A}_2\mathbf{B}_3 & \mathbf{A}_1\mathbf{B}_2 + \mathbf{A}_2\mathbf{B}_4 \\ \mathbf{A}_3\mathbf{B}_1 + \mathbf{A}_4\mathbf{B}_3 & \mathbf{A}_3\mathbf{B}_2 + \mathbf{A}_4\mathbf{B}_4 \end{bmatrix}.$$

Definition of Determinant

The *determinant* of a 2×2 matrix is given by the following scalar:

$$|\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}. \quad [\text{A.4.4}]$$

The determinant of an $n \times n$ matrix can be defined recursively. Let $\mathbf{A}_{ij}$ denote the $(n-1) \times (n-1)$ matrix formed by deleting row i and column j from $\mathbf{A}$. The determinant of $\mathbf{A}$ is given by

$$|\mathbf{A}| = \sum_{j=1}^n (-1)^{j+1} a_{1j} |\mathbf{A}_{1j}|. \quad [\text{A.4.5}]$$

For example, the determinant of a 3×3 matrix is

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Properties of Determinants

A square matrix is said to be *lower triangular* if all the elements above the principal diagonal are zero ($a_{ij} = 0$ for $j > i$):

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}.$$

The determinant of a lower triangular matrix is simply the product of the terms along the principal diagonal:

$$|\mathbf{A}| = a_{11}a_{22} \cdots a_{nn}. \quad [\text{A.4.6}]$$

That [A.4.6] holds for $n = 2$ follows immediately from [A.4.4]. Given that it holds for a matrix of order $n - 1$, equation [A.4.5] implies that it holds for n :

$$|\mathbf{A}| = a_{11} \begin{vmatrix} a_{22} & 0 & 0 & \cdots & 0 \\ a_{32} & a_{33} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{n2} & a_{n3} & a_{n4} & \cdots & a_{nn} \end{vmatrix} + 0 \cdot |\mathbf{A}_{12}| + \cdots + 0 \cdot |\mathbf{A}_{1n}|.$$

An immediate implication of [A.4.6] is that the determinant of the identity matrix is unity:

$$|\mathbf{I}_n| = 1. \quad [\text{A.4.7}]$$

Another useful fact about determinants is that if an $n \times n$ matrix $\mathbf{A}$ is multiplied by a scalar α , the effect is to multiply the determinant by α^n :

$$|\alpha\mathbf{A}| = \alpha^n |\mathbf{A}|. \quad [\text{A.4.8}]$$

Again, [A.4.8] is immediately apparent for the $n = 2$ case from [A.4.4]:

$$\begin{aligned} |\alpha\mathbf{A}| &= \begin{vmatrix} \alpha a_{11} & \alpha a_{12} \\ \alpha a_{21} & \alpha a_{22} \end{vmatrix} \\ &= (\alpha a_{11} \alpha a_{22}) - (\alpha a_{12} \alpha a_{21}) \\ &= \alpha^2 (a_{11} a_{22} - a_{12} a_{21}) \\ &= \alpha^2 |\mathbf{A}|. \end{aligned}$$

Given that it holds for $n - 1$, it is simple to verify for n using [A.4.5].

By contrast, if a single row of $\mathbf{A}$ is multiplied by the constant α (as opposed to multiplying the entire matrix by α), then the determinant is multiplied by α . If the row that is multiplied by α is the first row, then this result is immediately apparent from [A.4.5]. If only the i th row of $\mathbf{A}$ is multiplied by α , the result can be shown by recursively applying [A.4.5] until the elements of the i th row appear explicitly in the formula.

Suppose that some constant c times the second row of a 2×2 matrix is added to the first row. This operation has no effect on the determinant:

$$\begin{vmatrix} a_{11} + ca_{21} & a_{12} + ca_{22} \\ a_{21} & a_{22} \end{vmatrix} = (a_{11} + ca_{21})a_{22} - (a_{12} + ca_{22})a_{21} \\ = a_{11}a_{22} - a_{12}a_{21}.$$

Similarly, if some constant c times the third row of a 3×3 matrix is added to the

second row, the determinant will again be unchanged:

$$\begin{aligned}
 & \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} + ca_{31} & a_{22} + ca_{32} & a_{23} + ca_{33} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\
 &= a_{11} \begin{vmatrix} a_{22} + ca_{32} & a_{23} + ca_{33} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} + ca_{31} & a_{23} + ca_{33} \\ a_{31} & a_{33} \end{vmatrix} \\
 &\quad + a_{13} \begin{vmatrix} a_{21} + ca_{31} & a_{22} + ca_{32} \\ a_{31} & a_{32} \end{vmatrix} \\
 &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.
 \end{aligned}$$

In general, if any row of an $n \times n$ matrix is multiplied by c and added to another row, the new matrix will have the same determinant as the original. Similarly, multiplying any column by c and adding the result to another column will not change the determinant.

This can be viewed as a special case of the following result. If $\mathbf{A}$ and $\mathbf{B}$ are both $n \times n$ matrices, then

$$|\mathbf{AB}| = |\mathbf{A}| \cdot |\mathbf{B}|. \quad [\text{A.4.9}]$$

Adding c times the second column of a 2×2 matrix $\mathbf{A}$ to the first column can be thought of as postmultiplying $\mathbf{A}$ by the following matrix:

$$\mathbf{B} = \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}.$$

Since $\mathbf{B}$ is lower triangular with 1s along the principal diagonal, its determinant is unity, and so, from [A.4.9],

$$|\mathbf{AB}| = |\mathbf{A}|.$$

Thus, the fact that adding a multiple of one column to another does not alter the determinant can be viewed as an implication of [A.4.9].

If two rows of a matrix are switched, the determinant changes signs. To switch the i th row with the j th, multiply the i th row by -1 ; this changes the sign of the determinant. Then subtract row i from row j , add the new j back to i , and subtract i from j once again. These last operations complete the switch and do not affect the determinant further. For example, let $\mathbf{A}$ be a (4×4) matrix written in partitioned form as

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}'_1 \\ \mathbf{a}'_2 \\ \mathbf{a}'_3 \\ \mathbf{a}'_4 \end{bmatrix},$$

where the (1×4) vector $\mathbf{a}'_i$ represents the i th row of $\mathbf{A}$. The determinant when rows 1 and 4 are switched can be calculated from

$$\begin{vmatrix} \mathbf{a}'_1 \\ \mathbf{a}'_2 \\ \mathbf{a}'_3 \\ \mathbf{a}'_4 \end{vmatrix} = - \begin{vmatrix} -\mathbf{a}'_1 \\ \mathbf{a}'_2 \\ \mathbf{a}'_3 \\ \mathbf{a}'_4 \end{vmatrix} = - \begin{vmatrix} -\mathbf{a}'_1 \\ \mathbf{a}'_2 \\ \mathbf{a}'_3 \\ \mathbf{a}'_1 + \mathbf{a}'_4 \end{vmatrix} = - \begin{vmatrix} \mathbf{a}'_4 \\ \mathbf{a}'_2 \\ \mathbf{a}'_3 \\ \mathbf{a}'_1 + \mathbf{a}'_4 \end{vmatrix} = - \begin{vmatrix} \mathbf{a}'_4 \\ \mathbf{a}'_2 \\ \mathbf{a}'_3 \\ \mathbf{a}'_1 \end{vmatrix}.$$

This result permits calculation of the determinant of $\mathbf{A}$ in reference to any row of an $(n \times n)$ matrix $\mathbf{A}$:

$$|\mathbf{A}| = \sum_{j=1}^n (-1)^{i+j} a_{ij} |\mathbf{A}_{ij}|. \quad [\text{A.4.10}]$$

To derive [A.4.10], define $\mathbf{A}^*$ as

$$\mathbf{A}^* \equiv \begin{bmatrix} \mathbf{a}'_i \\ \mathbf{a}'_1 \\ \mathbf{a}'_2 \\ \vdots \\ \mathbf{a}'_{i-1} \\ \mathbf{a}'_{i+1} \\ \vdots \\ \mathbf{a}'_n \end{bmatrix}.$$

Then, from [A.4.5],

$$|\mathbf{A}^*| = \sum_{j=1}^n (-1)^{j+1} a_{1j}^* |\mathbf{A}_{1j}^*| = \sum_{j=1}^n (-1)^{j+1} a_{ij} |\mathbf{A}_{ij}|.$$

Moreover, $\mathbf{A}^*$ is obtained from $\mathbf{A}$ by $(i-1)$ row switches, such as switching i with $i-1$, $i-1$ with $i-2$, . . . , and 2 with 1. Hence,

$$|\mathbf{A}| = (-1)^{i-1} |\mathbf{A}^*| = (-1)^{i-1} \sum_{j=1}^n (-1)^{j+1} a_{ij} |\mathbf{A}_{ij}|,$$

as claimed in [A.4.10].

An immediate implication of [A.4.10] is that if any row of a matrix contains all zeros, then the determinant of the matrix is zero.

It can also be shown that the transpose of a matrix has the same determinant as the original matrix:

$$|\mathbf{A}'| = |\mathbf{A}|. \quad [\text{A.4.11}]$$

This means, for example, that if the k th column of a matrix consists entirely of zeros, then the determinant of the matrix is zero. It also implies that the determinant of an *upper triangular matrix* (one for which $a_{ij} = 0$ for all $j < i$) is the product of the terms on the principal diagonal.

Adjoint of a Matrix

Let $\mathbf{A}$ denote an $(n \times n)$ matrix, and as before let $\mathbf{A}_{ji}$ denote the $[(n-1) \times (n-1)]$ matrix that results from deleting row j and column i of $\mathbf{A}$. The *adjoint* of $\mathbf{A}$ is the $(n \times n)$ matrix whose row i , column j element is given by $(-1)^{i+j} |\mathbf{A}_{ji}|$.

Inverse of a Matrix

If the determinant of an $n \times n$ matrix $\mathbf{A}$ is not equal to zero, its *inverse* (an $n \times n$ matrix denoted $\mathbf{A}^{-1}$) exists and is found by dividing the adjoint by the determinant:

$$\mathbf{A}^{-1} = (1/|\mathbf{A}|) \cdot [(-1)^{i+j} |\mathbf{A}_{ji}|]. \quad [\text{A.4.12}]$$

For example, for $n = 2$,

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^{-1} = (1/[a_{11}a_{22} - a_{12}a_{21}]) \cdot \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}. \quad [\text{A.4.13}]$$

A matrix whose inverse exists is said to be *nonsingular*. A matrix whose determinant is zero is *singular* and has no inverse.

When an inverse exists,

$$\mathbf{A} \times \mathbf{A}^{-1} = \mathbf{I}_n. \quad [\text{A.4.14}]$$

Taking determinants of both sides of [A.4.14] and using [A.4.9] and [A.4.7],

$$|\mathbf{A}| \cdot |\mathbf{A}^{-1}| = 1,$$

so

$$|\mathbf{A}^{-1}| = 1/|\mathbf{A}|. \quad [\text{A.4.15}]$$

Alternatively, taking the transpose of both sides of [A.4.14] and recalling [A.4.3],

$$(\mathbf{A}^{-1})' \mathbf{A}' = \mathbf{I}_n,$$

which means that $(\mathbf{A}^{-1})'$ is the inverse of $\mathbf{A}'$:

$$(\mathbf{A}^{-1})' = (\mathbf{A}')^{-1}.$$

For α a nonzero scalar and $\mathbf{A}$ a nonsingular matrix,

$$[\alpha \mathbf{A}]^{-1} = \alpha^{-1} \mathbf{A}^{-1}.$$

Also, for $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ all nonsingular ($n \times n$) matrices,

$$[\mathbf{AB}]^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$$

and

$$[\mathbf{ABC}]^{-1} = \mathbf{C}^{-1} \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

Linear Dependence

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ be a set of k different ($n \times 1$) vectors. The vectors are said to be *linearly dependent* if there exists a set of k scalars ($c_1, c_2, \dots, c_k$), not all of which are zero, such that

$$c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2 + \dots + c_k \mathbf{x}_k = \mathbf{0}.$$

If no such set of nonzero numbers ($c_1, c_2, \dots, c_k$) exists, then the vectors ($\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$) are said to be *linearly independent*.

Suppose the vectors ($\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$) are collected in an ($n \times k$) matrix $\mathbf{T}$, written in partitioned form as

$$\mathbf{T} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_k].$$

If the number of vectors (k) is equal to the dimension of each vector (n), then there is a simple relation between the notion of linear dependence and the determinant of the ($n \times n$) matrix $\mathbf{T}$; specifically, if ($\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$) are linearly dependent, then $|\mathbf{T}| = 0$. To see this, suppose that $\mathbf{x}_1$ is one of the vectors that have a nonzero value of c_i . Then linear dependence means that

$$\mathbf{x}_1 = -(c_2/c_1)\mathbf{x}_2 - (c_3/c_1)\mathbf{x}_3 - \dots - (c_n/c_1)\mathbf{x}_n.$$

Then the determinant of $\mathbf{T}$ is equal to

$$|\mathbf{T}| = |[-(c_2/c_1)\mathbf{x}_2 - (c_3/c_1)\mathbf{x}_3 - \cdots - (c_n/c_1)\mathbf{x}_n] \quad \mathbf{x}_2 \quad \cdots \quad \mathbf{x}_n|.$$

But if we add (c_n/c_1) times the n th column to the first column, (c_{n-1}/c_1) times the $(n-1)$ th column to the first column, . . . , and (c_2/c_1) times the second column to the first column, the result is

$$\begin{aligned} |\mathbf{T}| &= |\mathbf{0} \quad \mathbf{x}_2 \quad \cdots \quad \mathbf{x}_n| \\ &= 0. \end{aligned}$$

The converse can also be shown to be true: if $|\mathbf{T}| = 0$, then $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ are linearly dependent.

Eigenvalues and Eigenvectors

Suppose that an $n \times n$ matrix $\mathbf{A}$, a nonzero $n \times 1$ vector $\mathbf{x}$, and a scalar λ are related by

$$\mathbf{Ax} = \lambda\mathbf{x}. \quad [\text{A.4.16}]$$

Then $\mathbf{x}$ is called an *eigenvector* of $\mathbf{A}$ and λ the associated *eigenvalue*. Equation [A.4.16] can be written

$$\mathbf{Ax} - \lambda\mathbf{I}_n\mathbf{x} = \mathbf{0}$$

or

$$(\mathbf{A} - \lambda\mathbf{I}_n)\mathbf{x} = \mathbf{0}. \quad [\text{A.4.17}]$$

Suppose that the matrix $(\mathbf{A} - \lambda\mathbf{I}_n)$ were nonsingular. Then $(\mathbf{A} - \lambda\mathbf{I}_n)^{-1}$ would exist and we could premultiply [A.4.17] by $(\mathbf{A} - \lambda\mathbf{I}_n)^{-1}$ to deduce that

$$\mathbf{x} = \mathbf{0}.$$

Thus, if a nonzero vector $\mathbf{x}$ exists that satisfies [A.4.16], then it must be associated with a value of λ such that $(\mathbf{A} - \lambda\mathbf{I}_n)$ is singular. An eigenvalue of the matrix $\mathbf{A}$ is therefore a number λ such that

$$|\mathbf{A} - \lambda\mathbf{I}_n| = 0. \quad [\text{A.4.18}]$$

Eigenvalues of Triangular Matrices

Notice that if $\mathbf{A}$ is upper triangular or lower triangular, then $\mathbf{A} - \lambda\mathbf{I}_n$ is as well, and its determinant is just the product of terms along the principal diagonal:

$$|\mathbf{A} - \lambda\mathbf{I}_n| = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda).$$

Thus, for a triangular matrix, the eigenvalues (the values of λ for which this expression equals zero) are just the values of $\mathbf{A}$ along the principal diagonal.

Linear Independence of Eigenvectors

A useful result is that if the eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_n)$ are all distinct, then the associated eigenvectors $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ are linearly independent. To see this for the case $n = 2$, consider any numbers c_1 and c_2 such that

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 = \mathbf{0}. \quad [\text{A.4.19}]$$

Premultiplying both sides of [A.4.19] by $\mathbf{A}$ produces

$$c_1 \mathbf{A} \mathbf{x}_1 + c_2 \mathbf{A} \mathbf{x}_2 = c_1 \lambda_1 \mathbf{x}_1 + c_2 \lambda_2 \mathbf{x}_2 = \mathbf{0}. \quad [\text{A.4.20}]$$

If [A.4.19] is multiplied by λ_1 and subtracted from [A.4.20], the result is

$$c_2(\lambda_2 - \lambda_1) \mathbf{x}_2 = \mathbf{0}. \quad [\text{A.4.21}]$$

But $\mathbf{x}_2$ is an eigenvector of $\mathbf{A}$, and so it cannot be the zero vector. Also, $\lambda_2 - \lambda_1$ cannot be zero, since $\lambda_2 \neq \lambda_1$. Equation [A.4.21] therefore implies that $c_2 = 0$. A parallel set of calculations show that $c_1 = 0$. Thus, the only values of c_1 and c_2 consistent with [A.4.19] are $c_1 = 0$ and $c_2 = 0$, which means that $\mathbf{x}_1$ and $\mathbf{x}_2$ are linearly independent. A similar argument for $n > 2$ can be made by induction.

A Useful Decomposition

Suppose an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_n)$. Collect these in a diagonal matrix $\mathbf{\Lambda}$:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Collect the eigenvectors $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ in an $(n \times n)$ matrix $\mathbf{T}$:

$$\mathbf{T} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \cdots \quad \mathbf{x}_n].$$

Applying the formula for multiplying partitioned matrices,

$$\mathbf{AT} = [\mathbf{Ax}_1 \quad \mathbf{Ax}_2 \quad \cdots \quad \mathbf{Ax}_n].$$

But since $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ are eigenvectors, equation [A.4.16] implies that

$$\mathbf{AT} = [\lambda_1 \mathbf{x}_1 \quad \lambda_2 \mathbf{x}_2 \quad \cdots \quad \lambda_n \mathbf{x}_n]. \quad [\text{A.4.22}]$$

A second application of the formula for multiplying partitioned matrices shows that the right side of [A.4.22] is in turn equal to

$$\begin{aligned} & [\lambda_1 \mathbf{x}_1 \quad \lambda_2 \mathbf{x}_2 \quad \cdots \quad \lambda_n \mathbf{x}_n] \\ &= [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \cdots \quad \mathbf{x}_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \\ &= \mathbf{TA}. \end{aligned}$$

Thus, [A.4.22] can be written

$$\mathbf{AT} = \mathbf{TA}. \quad [\text{A.4.23}]$$

Now, since the eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_n)$ are taken to be distinct, the eigenvectors $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ are known to be linearly independent. Thus, $|\mathbf{T}| \neq 0$ and $\mathbf{T}^{-1}$ exists. Postmultiplying [A.4.23] by $\mathbf{T}^{-1}$ reveals a useful decomposition of $\mathbf{A}$:

$$\mathbf{A} = \mathbf{TAT}^{-1}. \quad [\text{A.4.24}]$$

The Jordan Decomposition

The decomposition in [A.4.24] required the $(n \times n)$ matrix $\mathbf{A}$ to have n linearly independent eigenvectors. This will be true whenever $\mathbf{A}$ has n distinct

eigenvalues, and could still be true even if $\mathbf{A}$ has some repeated eigenvalues. In the completely general case when $\mathbf{A}$ has $s \leq n$ linearly independent eigenvectors, there always exists a decomposition similar to [A.4.24], known as the *Jordan decomposition*. Specifically, for such a matrix $\mathbf{A}$ there exists a nonsingular $(n \times n)$ matrix $\mathbf{M}$ such that

$$\mathbf{A} = \mathbf{M}\mathbf{J}\mathbf{M}^{-1}, \quad [\text{A.4.25}]$$

where the $(n \times n)$ matrix $\mathbf{J}$ takes the form

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{J}_s \end{bmatrix} \quad [\text{A.4.26}]$$

with

$$\mathbf{J}_i = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 \\ 0 & 0 & \lambda_i & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_i \end{bmatrix}. \quad [\text{A.4.27}]$$

Thus, $\mathbf{J}_i$ has the eigenvalue λ_i repeated along the principal diagonal and has unity repeated along the diagonal above the principal diagonal. The same eigenvalue λ_i can appear in two different Jordan blocks $\mathbf{J}_i$ and $\mathbf{J}_k$ if it corresponds to several linearly independent eigenvectors.

Some Further Results on Eigenvalues

Suppose that λ is an eigenvalue of the $(n \times n)$ matrix $\mathbf{A}$. Then λ is also an eigenvalue of $\mathbf{SAS}^{-1}$ for any nonsingular $(n \times n)$ matrix $\mathbf{S}$. To see this, note that

$$(\mathbf{A} - \lambda \mathbf{I}_n)\mathbf{x} = \mathbf{0}$$

implies that

$$\mathbf{S}(\mathbf{A} - \lambda \mathbf{I}_n)\mathbf{S}^{-1}\mathbf{Sx} = \mathbf{0}$$

or

$$(\mathbf{SAS}^{-1} - \lambda \mathbf{I}_n)\mathbf{x}^* = \mathbf{0} \quad [\text{A.4.28}]$$

for $\mathbf{x}^* \equiv \mathbf{Sx}$. Thus, λ is an eigenvalue of $\mathbf{SAS}^{-1}$ associated with the eigenvector $\mathbf{x}^*$.

From [A.4.25], this implies that the determinant of any $(n \times n)$ matrix $\mathbf{A}$ is the same as the determinant of its Jordan matrix $\mathbf{J}$ defined in [A.4.26]. Since $\mathbf{J}$ is upper triangular, its determinant is the product of terms along the principal diagonal, which were just the eigenvalues of $\mathbf{A}$. Thus, the determinant of any matrix $\mathbf{A}$ is given by the product of its eigenvalues.

It is also clear that the eigenvalues of $\mathbf{A}$ are the same as those of $\mathbf{A}'$. Taking the transpose of [A.4.25],

$$\mathbf{A}' = (\mathbf{M}')^{-1}\mathbf{J}'\mathbf{M}',$$

we see that the eigenvalues of $\mathbf{A}'$ are the eigenvalues of $\mathbf{J}'$. Since $\mathbf{J}'$ is lower

triangular, its eigenvalues are the elements on its principal diagonal. But $\mathbf{J}'$ and $\mathbf{J}$ have the same principal diagonal, meaning that $\mathbf{A}'$ and $\mathbf{A}$ have the same eigenvalues.

Matrix Geometric Series

The results of [A.3.6] through [A.3.10] generalize readily to geometric series involving square matrices. Consider the sum

$$\mathbf{S}_T = \mathbf{I}_n + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots + \mathbf{A}^T \quad [\text{A.4.29}]$$

for $\mathbf{A}$ an $(n \times n)$ matrix. Premultiplying both sides of [A.4.29] by $\mathbf{A}$, we see that

$$\mathbf{A}\mathbf{S}_T = \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots + \mathbf{A}^T + \mathbf{A}^{T+1}. \quad [\text{A.4.30}]$$

Subtracting [A.4.30] from [A.4.29], we find that

$$(\mathbf{I}_n - \mathbf{A})\mathbf{S}_T = \mathbf{I}_n - \mathbf{A}^{T+1}. \quad [\text{A.4.31}]$$

Notice from [A.4.18] that if $|\mathbf{I}_n - \mathbf{A}| = 0$, then $\lambda = 1$ would be an eigenvalue of $\mathbf{A}$. Assuming that none of the eigenvalues of $\mathbf{A}$ is equal to unity, the matrix $(\mathbf{I}_n - \mathbf{A})$ is nonsingular and [A.4.31] implies that

$$\mathbf{S}_T = (\mathbf{I}_n - \mathbf{A})^{-1}(\mathbf{I}_n - \mathbf{A}^{T+1}) \quad [\text{A.4.32}]$$

if no eigenvalue of $\mathbf{A}$ equals 1. If all the eigenvalues of $\mathbf{A}$ are strictly less than 1 in modulus, it can be shown that $\mathbf{A}^{T+1} \rightarrow \mathbf{0}$ as $T \rightarrow \infty$, implying that

$$(\mathbf{I}_n + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \cdots) = (\mathbf{I}_n - \mathbf{A})^{-1} \quad [\text{A.4.33}]$$

assuming that the eigenvalues of $\mathbf{A}$ are all inside unit circle.

Kronecker Products

For $\mathbf{A}$ an $(m \times n)$ matrix and $\mathbf{B}$ a $(p \times q)$ matrix, the *Kronecker product* of $\mathbf{A}$ and $\mathbf{B}$ is defined as the following $(mp) \times (nq)$ matrix:

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2n}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}.$$

The following properties of the Kronecker product are readily verified. For any matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$,

$$(\mathbf{A} \otimes \mathbf{B})' = \mathbf{A}' \otimes \mathbf{B}' \quad [\text{A.4.34}]$$

$$(\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C}). \quad [\text{A.4.35}]$$

Also, for $\mathbf{A}$ and $\mathbf{B}$ both $(m \times n)$ matrices and $\mathbf{C}$ any matrix,

$$(\mathbf{A} + \mathbf{B}) \otimes \mathbf{C} = (\mathbf{A} \otimes \mathbf{C}) + (\mathbf{B} \otimes \mathbf{C}) \quad [\text{A.4.36}]$$

$$\mathbf{C} \otimes (\mathbf{A} + \mathbf{B}) = (\mathbf{C} \otimes \mathbf{A}) + (\mathbf{C} \otimes \mathbf{B}). \quad [\text{A.4.37}]$$

Let $\mathbf{A}$ be $(m \times n)$, $\mathbf{B}$ be $(p \times q)$, $\mathbf{C}$ be $(n \times k)$, and $\mathbf{D}$ be $(q \times r)$. Then

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD}); \quad [\text{A.4.38}]$$

that is,

$$\begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2n}\mathbf{B} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix} \begin{bmatrix} c_{11}\mathbf{D} & c_{12}\mathbf{D} & \cdots & c_{1k}\mathbf{D} \\ c_{21}\mathbf{D} & c_{22}\mathbf{D} & \cdots & c_{2k}\mathbf{D} \\ \vdots & \vdots & \cdots & \vdots \\ c_{n1}\mathbf{D} & c_{n2}\mathbf{D} & \cdots & c_{nk}\mathbf{D} \end{bmatrix} \\ = \begin{bmatrix} \sum a_{1j}c_{j1}\mathbf{B}\mathbf{D} & \sum a_{1j}c_{j2}\mathbf{B}\mathbf{D} & \cdots & \sum a_{1j}c_{jk}\mathbf{B}\mathbf{D} \\ \sum a_{2j}c_{j1}\mathbf{B}\mathbf{D} & \sum a_{2j}c_{j2}\mathbf{B}\mathbf{D} & \cdots & \sum a_{2j}c_{jk}\mathbf{B}\mathbf{D} \\ \vdots & \vdots & \cdots & \vdots \\ \sum a_{mj}c_{j1}\mathbf{B}\mathbf{D} & \sum a_{mj}c_{j2}\mathbf{B}\mathbf{D} & \cdots & \sum a_{mj}c_{jk}\mathbf{B}\mathbf{D} \end{bmatrix}.$$

For $\mathbf{A}$ ($n \times n$) and $\mathbf{B}$ ($p \times p$) both nonsingular matrices we can set $\mathbf{C} = \mathbf{A}^{-1}$ and $\mathbf{D} = \mathbf{B}^{-1}$ in [A.4.38] to deduce that

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^{-1} \otimes \mathbf{B}^{-1}) = (\mathbf{A}\mathbf{A}^{-1}) \otimes (\mathbf{B}\mathbf{B}^{-1}) = \mathbf{I}_n \otimes \mathbf{I}_p = \mathbf{I}_{np}.$$

Thus,

$$(\mathbf{A} \otimes \mathbf{B})^{-1} = (\mathbf{A}^{-1} \otimes \mathbf{B}^{-1}). \quad [\text{A.4.39}]$$

Eigenvalues of a Kronecker Product

For $\mathbf{A}$ an ($n \times n$) matrix with (possibly nondistinct) eigenvalues ($\lambda_1, \lambda_2, \dots, \lambda_n$) and $\mathbf{B}$ ($p \times p$) with eigenvalues ($\mu_1, \mu_2, \dots, \mu_p$), then the (np) eigenvalues of $\mathbf{A} \otimes \mathbf{B}$ are given by $\lambda_i \mu_j$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, p$. To see this, write $\mathbf{A}$ and $\mathbf{B}$ in Jordan form as

$$\begin{aligned} \mathbf{A} &= \mathbf{M}_A \mathbf{J}_A \mathbf{M}_A^{-1} \\ \mathbf{B} &= \mathbf{M}_B \mathbf{J}_B \mathbf{M}_B^{-1}. \end{aligned}$$

Then $(\mathbf{M}_A \otimes \mathbf{M}_B)$ has inverse given by $(\mathbf{M}_A^{-1} \otimes \mathbf{M}_B^{-1})$. Moreover, we know from [A.4.28] that the eigenvalues of $(\mathbf{A} \otimes \mathbf{B})$ are the same as the eigenvalues of

$$\begin{aligned} (\mathbf{M}_A^{-1} \otimes \mathbf{M}_B^{-1})(\mathbf{A} \otimes \mathbf{B})(\mathbf{M}_A \otimes \mathbf{M}_B) &= (\mathbf{M}_A^{-1} \mathbf{A} \mathbf{M}_A) \otimes (\mathbf{M}_B^{-1} \mathbf{B} \mathbf{M}_B) \\ &= \mathbf{J}_A \otimes \mathbf{J}_B. \end{aligned}$$

But $\mathbf{J}_A$ and $\mathbf{J}_B$ are both upper triangular, meaning that $(\mathbf{J}_A \otimes \mathbf{J}_B)$ is upper triangular as well. The eigenvalues of $(\mathbf{A} \otimes \mathbf{B})$ are thus just the terms on the principal diagonal of $(\mathbf{J}_A \otimes \mathbf{J}_B)$, which are given by $\lambda_i \mu_j$.

Positive Definite Matrices

An ($n \times n$) real symmetric matrix $\mathbf{A}$ is said to be *positive semidefinite* if for any real ($n \times 1$) vector $\mathbf{x}$,

$$\mathbf{x}'\mathbf{A}\mathbf{x} \geq 0.$$

We make the stronger statement that a real symmetric matrix $\mathbf{A}$ is *positive definite* if for any real nonzero ($n \times 1$) vector $\mathbf{x}$,

$$\mathbf{x}'\mathbf{A}\mathbf{x} > 0;$$

hence, any positive definite matrix could also be said to be positive semidefinite.

Let λ be an eigenvalue of $\mathbf{A}$ associated with the eigenvector $\mathbf{x}$:

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$

Premultiplying this equation by $\mathbf{x}'$ results in

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \lambda\mathbf{x}'\mathbf{x}.$$

Since an eigenvector $\mathbf{x}$ cannot be the zero vector, $\mathbf{x}'\mathbf{x} > 0$. Thus, for a positive semidefinite matrix $\mathbf{A}$, any eigenvalue λ of $\mathbf{A}$ must be greater than or equal to zero. For $\mathbf{A}$ positive definite, all eigenvalues are strictly greater than zero. Since the determinant of $\mathbf{A}$ is the product of the eigenvalues, the determinant of a positive definite matrix $\mathbf{A}$ is strictly positive.

Let $\mathbf{A}$ be a positive definite ($n \times n$) matrix and let $\mathbf{B}$ denote a nonsingular ($n \times n$) matrix. Then $\mathbf{B}'\mathbf{A}\mathbf{B}$ is positive definite. To see this, let $\mathbf{x}$ be any nonzero vector. Define

$$\tilde{\mathbf{x}} \equiv \mathbf{B}\mathbf{x}.$$

Then $\tilde{\mathbf{x}}$ cannot be the zero vector, for if it were, this equation would state that there exists a nonzero vector $\mathbf{x}$ such that

$$\mathbf{B}\mathbf{x} = \mathbf{0},$$

in which case zero would be an eigenvalue of $\mathbf{B}$ associated with the eigenvector $\mathbf{x}$. But since $\mathbf{B}$ is nonsingular, none of its eigenvalues can be zero. Thus, $\tilde{\mathbf{x}} = \mathbf{B}\mathbf{x}$ cannot be the zero vector, and

$$\mathbf{x}'\mathbf{B}'\mathbf{A}\mathbf{B}\mathbf{x} = \tilde{\mathbf{x}}'\mathbf{A}\tilde{\mathbf{x}} > 0,$$

establishing that the matrix $\mathbf{B}'\mathbf{A}\mathbf{B}$ is positive definite.

A special case of this result is obtained by letting $\mathbf{A}$ be the identity matrix. Then the result implies that any matrix that can be written as $\mathbf{B}'\mathbf{B}$ for some nonsingular matrix $\mathbf{B}$ is positive definite. More generally, any matrix that can be written as $\mathbf{B}'\mathbf{B}$ for an arbitrary matrix $\mathbf{B}$ must be positive semidefinite:

$$\mathbf{x}'\mathbf{B}'\mathbf{B}\mathbf{x} = \tilde{\mathbf{x}}'\tilde{\mathbf{x}} = \tilde{x}_1^2 + \tilde{x}_2^2 + \cdots + \tilde{x}_n^2 \geq 0, \quad [\text{A.4.40}]$$

where $\tilde{\mathbf{x}} \equiv \mathbf{B}\mathbf{x}$.

The converse propositions are also true: if $\mathbf{A}$ is positive semidefinite, then there exists a matrix $\mathbf{B}$ such that $\mathbf{A} = \mathbf{B}'\mathbf{B}$; if $\mathbf{A}$ is positive definite, then there exists a nonsingular matrix $\mathbf{B}$ such that $\mathbf{A} = \mathbf{B}'\mathbf{B}$. A proof of this claim and an algorithm for calculating $\mathbf{B}$ are provided in Section 4.4.

Conjugate Transposes

Let $\mathbf{A}$ denote an ($m \times n$) matrix of (possibly) complex numbers:

$$\mathbf{A} = \begin{bmatrix} a_{11} + b_{11}i & \cdots & a_{1n} + b_{1n}i \\ a_{21} + b_{21}i & \cdots & a_{2n} + b_{2n}i \\ \vdots & \cdots & \vdots \\ a_{m1} + b_{m1}i & \cdots & a_{mn} + b_{mn}i \end{bmatrix}.$$

The *conjugate transpose* of $\mathbf{A}$, denoted $\mathbf{A}^H$, is formed by transposing $\mathbf{A}$ and replacing each element with its complex conjugate:

$$\mathbf{A}^H = \begin{bmatrix} a_{11} - b_{11}i & \cdots & a_{m1} - b_{m1}i \\ a_{12} - b_{12}i & \cdots & a_{m2} - b_{m2}i \\ \vdots & \cdots & \vdots \\ a_{1n} - b_{1n}i & \cdots & a_{mn} - b_{mn}i \end{bmatrix}.$$

Thus, if $\mathbf{A}$ is real, then $\mathbf{A}^H$ and $\mathbf{A}'$ would denote the same matrix.

Notice that if an $(n \times 1)$ complex vector is premultiplied by its conjugate transpose, the result is a nonnegative real scalar:

$$\begin{aligned}\mathbf{x}^H \mathbf{x} &= [(a_1 - b_1 i) \quad (a_2 - b_2 i) \quad \cdots \quad (a_n - b_n i)] \begin{bmatrix} a_1 + b_1 i \\ a_2 + b_2 i \\ \vdots \\ a_n + b_n i \end{bmatrix} \\ &= \sum_{i=1}^n (a_i^2 + b_i^2) \geq 0.\end{aligned}$$

For $\mathbf{B}$ a real $(m \times n)$ matrix and $\mathbf{x}$ a complex $(n \times 1)$ vector,

$$(\mathbf{B}\mathbf{x})^H = \mathbf{x}^H \mathbf{B}'.$$

More generally, if both $\mathbf{B}$ and $\mathbf{x}$ are complex,

$$(\mathbf{B}\mathbf{x})^H = \mathbf{x}^H \mathbf{B}^H.$$

Notice that if $\mathbf{A}$ is positive semidefinite, then

$$\mathbf{x}^H \mathbf{A} \mathbf{x} = \mathbf{x}^H \mathbf{B}' \mathbf{B} \mathbf{x} = \tilde{\mathbf{x}}^H \tilde{\mathbf{x}},$$

with $\tilde{\mathbf{x}} \equiv \mathbf{B}\mathbf{x}$. Thus, $\mathbf{x}^H \mathbf{A} \mathbf{x}$ is a nonnegative real scalar for any $\mathbf{x}$ when $\mathbf{A}$ is positive semidefinite. It is a positive real scalar for $\mathbf{A}$ positive definite.

Continuity of Functions of Vectors

A function of more than one argument, such as

$$y = f(x_1, x_2, \dots, x_n), \quad [\text{A.4.41}]$$

is said to be *continuous* at $(c_1, c_2, \dots, c_n)$ if $f(c_1, c_2, \dots, c_n)$ is finite and for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x_1, x_2, \dots, x_n) - f(c_1, c_2, \dots, c_n)| < \varepsilon$$

whenever

$$(x_1 - c_1)^2 + (x_2 - c_2)^2 + \cdots + (x_n - c_n)^2 < \delta^2.$$

Partial Derivatives

The *partial derivative* of f with respect to x_i is defined by

$$\begin{aligned}\frac{\partial f}{\partial x_i} &= \lim_{\Delta \rightarrow 0} \Delta^{-1} \cdot \{f(x_1, x_2, \dots, x_{i-1}, x_i + \Delta, x_{i+1}, \dots, x_n) \\ &\quad - f(x_1, x_2, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n)\}.\end{aligned} \quad [\text{A.4.42}]$$

Gradient

If we collect the n partial derivatives in [A.4.42] in a vector, we obtain the *gradient* of the function f , denoted ∇ :

$$\underset{(n \times 1)}{\nabla} \equiv \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \\ \vdots \\ \partial f / \partial x_n \end{bmatrix}. \quad [\text{A.4.43}]$$

For example, suppose f is a linear function:

$$f(x_1, x_2, \dots, x_n) = a_1x_1 + a_2x_2 + \dots + a_nx_n. \quad [\text{A.4.44}]$$

Define $\mathbf{a}$ and $\mathbf{x}$ to be the following $(n \times 1)$ vectors:

$$\mathbf{a} \equiv \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad [\text{A.4.45}]$$

$$\mathbf{x} \equiv \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}. \quad [\text{A.4.46}]$$

Then [A.4.44] can be written

$$f(\mathbf{x}) = \mathbf{a}'\mathbf{x}.$$

The partial derivative of $f(\cdot)$ with respect to the i th argument is

$$\frac{\partial f}{\partial x_i} = a_i,$$

and the gradient is

$$\nabla = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \mathbf{a}.$$

Second-Order Derivatives

A *second-order derivative* of [A.4.41] is given by

$$\frac{\partial^2 f(x_1, \dots, x_n)}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left[\frac{\partial f(x_1, \dots, x_n)}{\partial x_j} \right].$$

Where second-order derivatives exist and are continuous for all i and j , the order of differentiation is irrelevant:

$$\frac{\partial}{\partial x_i} \left[\frac{\partial f(x_1, \dots, x_n)}{\partial x_j} \right] = \frac{\partial}{\partial x_j} \left[\frac{\partial f(x_1, \dots, x_n)}{\partial x_i} \right].$$

Sometimes these second-order derivatives are collected in an $n \times n$ matrix $\mathbf{H}$ called the *Hessian matrix*:

$$\mathbf{H} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right].$$

We will also use the notation

$$\frac{\partial^2 f}{\partial \mathbf{x} \partial \mathbf{x}'}$$

to represent the matrix $\mathbf{H}$.

Derivatives of Vector-Valued Functions

Suppose we have a set of m functions $f_1(\cdot), f_2(\cdot), \dots, f_m(\cdot)$, each of which depends on the n variables $(x_1, x_2, \dots, x_n)$. We can collect the m functions into a single vector-valued function:

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix}.$$

We sometimes write

$$\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

to indicate that the function takes n different real numbers (summarized by the vector $\mathbf{x}$, an element of $\mathbb{R}^n$) and calculates m different new numbers (summarized by the value of $\mathbf{f}$, an element of $\mathbb{R}^m$). Suppose that each of the functions $f_1(\cdot), f_2(\cdot), \dots, f_m(\cdot)$ has derivatives with respect to each of the arguments $x_1, x_2, \dots, x_n$. We can summarize these derivatives in an $(m \times n)$ matrix, called the *Jacobian matrix* of $\mathbf{f}$ and indicated by $\partial \mathbf{f} / \partial \mathbf{x}'$:

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}'} \equiv \begin{bmatrix} \partial f_1 / \partial x_1 & \partial f_1 / \partial x_2 & \cdots & \partial f_1 / \partial x_n \\ \partial f_2 / \partial x_1 & \partial f_2 / \partial x_2 & \cdots & \partial f_2 / \partial x_n \\ \vdots & \vdots & \cdots & \vdots \\ \partial f_m / \partial x_1 & \partial f_m / \partial x_2 & \cdots & \partial f_m / \partial x_n \end{bmatrix}.$$

For example, suppose that each of the functions $f_i(\mathbf{x})$ is linear:

$$f_1(\mathbf{x}) = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n$$

$$f_2(\mathbf{x}) = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n$$

$$\vdots$$

$$f_m(\mathbf{x}) = a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n.$$

We could write this system in matrix form as

$$\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x},$$

where

$$\mathbf{A} \equiv \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

and $\mathbf{x}$ is the $(n \times 1)$ vector defined in [A.4.46]. Then

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}'} = \mathbf{A}.$$

Taylor's Theorem with Multiple Arguments

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^1$ as in [A.4.41], with continuous second derivatives. A first-order Taylor series expansion of $f(\mathbf{x})$ around $\mathbf{c}$ is given by

$$f(\mathbf{x}) = f(\mathbf{c}) + \left. \frac{\partial f}{\partial \mathbf{x}'} \right|_{\mathbf{x}=\mathbf{c}} \cdot (\mathbf{x} - \mathbf{c}) + R_1(\mathbf{c}, \mathbf{x}). \quad [\text{A.4.47}]$$

Here $\partial f / \partial \mathbf{x}'$ denotes the $(1 \times n)$ vector that is the transpose of the gradient, and the remainder $R_1(\cdot)$ satisfies

$$R_1(\mathbf{c}, \mathbf{x}) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j} \bigg|_{\mathbf{x}=\delta(i,j)} \cdot (x_i - c_i)(x_j - c_j)$$

for $\delta(i, j)$ an $(n \times 1)$ vector, potentially different for each i and j , with each $\delta(i, j)$ between $\mathbf{c}$ and $\mathbf{x}$, that is, $\delta(i, j) = \lambda(i, j)\mathbf{c} + [1 - \lambda(i, j)]\mathbf{x}$ for some $\lambda(i, j)$ between 0 and 1. Furthermore,

$$\lim_{\mathbf{x} \rightarrow \mathbf{c}} \frac{R_1(\mathbf{c}, \mathbf{x})}{[(\mathbf{x} - \mathbf{c})'(\mathbf{x} - \mathbf{c})]^{1/2}} = 0.$$

An implication of [A.4.47] is that if we wish to approximate the consequences for f of simultaneously changing x_1 by Δ_1 , x_2 by Δ_2 , . . . , and x_n by Δ_n , we could use

$$\begin{aligned} f(x_1 + \Delta_1, x_2 + \Delta_2, \dots, x_n + \Delta_n) - f(x_1, x_2, \dots, x_n) \\ \cong \frac{\partial f}{\partial x_1} \cdot \Delta_1 + \frac{\partial f}{\partial x_2} \cdot \Delta_2 + \dots + \frac{\partial f}{\partial x_n} \cdot \Delta_n. \end{aligned} \quad [\text{A.4.48}]$$

If $f(\cdot)$ has continuous third derivatives, a second-order Taylor series expansion of $f(\mathbf{x})$ around $\mathbf{c}$ is given by

$$\begin{aligned} f(\mathbf{x}) = f(\mathbf{c}) + \frac{\partial f}{\partial \mathbf{x}'} \bigg|_{\mathbf{x}=\mathbf{c}} \cdot (\mathbf{x} - \mathbf{c}) \\ + \frac{1}{2} (\mathbf{x} - \mathbf{c})' \frac{\partial^2 f}{\partial \mathbf{x} \partial \mathbf{x}'} \bigg|_{\mathbf{x}=\mathbf{c}} (\mathbf{x} - \mathbf{c}) + R_2(\mathbf{c}, \mathbf{x}), \end{aligned} \quad [\text{A.4.49}]$$

where

$$R_2(\mathbf{c}, \mathbf{x}) = \frac{1}{3!} \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k} \bigg|_{\mathbf{x}=\delta(i,j,k)} \cdot (x_i - c_i)(x_j - c_j)(x_k - c_k)$$

with $\delta(i, j, k)$ between $\mathbf{c}$ and $\mathbf{x}$ and

$$\lim_{\mathbf{x} \rightarrow \mathbf{c}} \frac{R_2(\mathbf{c}, \mathbf{x})}{(\mathbf{x} - \mathbf{c})'(\mathbf{x} - \mathbf{c})} = 0.$$

Multiple Integrals

The notation

$$\int_a^b \int_c^d f(x, y) dy dx$$

indicates the following operation: first integrate

$$\int_c^d f(x, y) dy$$

with respect to y , with x held fixed, and then integrate the resulting function with respect to x . For example,

$$\int_0^1 \int_0^2 x^4 y dy dx = \int_0^1 x^4 [(2^2/2) - (0^2/2)] dx = 2[1^5/5 - 0^5/5] = 2/5.$$

Provided that $f(x, y)$ is continuous, the order of integration can be reversed. For example,

$$\int_0^2 \int_0^1 x^4 y \, dx \, dy = \int_0^2 (1^5/5)y \, dy = (1/5) \cdot (2^2/2) = 2/5.$$

A.5. Probability and Statistics

Densities and Distributions

A *stochastic* or *random* variable X is said to be *discrete-valued* if it can take on only one of K particular values; call these $x_1, x_2, \dots, x_K$. Its *probability distribution* is a set of numbers that give the probability of each outcome:

$$P\{X = x_k\} \equiv \text{probability that } X \text{ takes on the value } x_k, \quad k = 1, \dots, K.$$

The probabilities sum to unity:

$$\sum_{k=1}^K P\{X = x_k\} = 1.$$

Assuming that the possible outcomes are ordered $x_1 < x_2 < \dots < x_K$, the probability that X takes on a value less than or equal to the value x_j is given by

$$P\{X \leq x_j\} = \sum_{k=1}^j P\{X = x_k\}.$$

If X is equal to a constant c with probability 1, then X is *nonstochastic*.

The probability law for a *continuous-valued* random variable X can often be described by the *density* function $f_X(x)$ with

$$\int_{-\infty}^{\infty} f_X(x) \, dx = 1. \quad [\text{A.5.1}]$$

The subscript X in $f_X(x)$ indicates that this is the density of the random variable X ; the argument x of $f_X(x)$ indexes the integration in [A.5.1]. The *cumulative distribution function* of X (denoted $F_X(a)$) gives the probability that X takes on a value less than or equal to a :

$$\begin{aligned} F_X(a) &\equiv P\{X \leq a\} \\ &= \int_{-\infty}^a f_X(x) \, dx. \end{aligned} \quad [\text{A.5.2}]$$

Population Moments

The *population mean* μ of a continuous-valued random variable X is given by

$$\mu = \int_{-\infty}^{\infty} x \cdot f_X(x) \, dx,$$

provided this integral exists. (In the formulas that follow, we assume for simplicity of exposition that the density functions are continuous and that the indicated

integrals all exist.) The *population variance* is

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f_X(x) \, dx.$$

The square root of the variance is called the *population standard deviation*.

In general, the *rth population moment* is given by

$$\int_{-\infty}^{\infty} x^r \cdot f_X(x) \, dx.$$

The population mean could thus be described as the first population moment.

Expectation

The population mean μ is also called the *expectation* of X , denoted $E(X)$ or sometimes simply EX . In general, the expectation of a function $g(X)$ is given by

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) \, dx, \quad [\text{A.5.3}]$$

where $f_X(x)$ is the density of X . For example, the *rth* population moment of X is the expectation of X^r .

Consider the random variable $a + bX$ for constants a and b . Its expectation is

$$\begin{aligned} E(a + bX) &= \int_{-\infty}^{\infty} [a + bx] \cdot f_X(x) \, dx \\ &= a \int_{-\infty}^{\infty} f_X(x) \, dx + b \int_{-\infty}^{\infty} x \cdot f_X(x) \, dx \\ &= a + b \cdot E(X). \end{aligned}$$

The variance of $a + bX$ is

$$\begin{aligned} \text{Var}(a + bX) &= \int_{-\infty}^{\infty} [(a + bx) - (a + b\mu)]^2 \cdot f_X(x) \, dx \\ &= b^2 \cdot \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f_X(x) \, dx \\ &= b^2 \cdot \text{Var}(X). \end{aligned} \quad [\text{A.5.4}]$$

Another useful result is

$$\begin{aligned} E(X^2) &= E[(X - \mu + \mu)^2] \\ &= E[(X - \mu)^2 + 2\mu(X - \mu) + \mu^2] \\ &= E[(X - \mu)^2] + 2\mu \cdot [E(X) - \mu] + \mu^2 \\ &= \text{Var}(X) + 0 + [E(X)]^2. \end{aligned}$$

To simplify the appearance of expressions, we adopt the convention that exponentiation and multiplication are carried out before the expectation operator. Thus, we will use $E(X - \mu + \mu)^2$ to indicate the same operation as $E[(X - \mu + \mu)^2]$. The square of $E(X - \mu + \mu)$ is indicated by using additional parentheses, as $[E(X - \mu + \mu)]^2$.

Sample Moments

A *sample moment* is a particular estimate of a population moment based on an observed set of data, say, $\{x_1, x_2, \dots, x_T\}$. The first sample moment is the

sample mean,

$$\bar{x} \equiv (1/T) \cdot (x_1 + x_2 + \cdots + x_T),$$

which is a natural estimate of the population mean μ . The sample variance,

$$s^2 \equiv (1/T) \cdot \sum_{t=1}^T (x_t - \bar{x})^2,$$

affords an estimate of the population variance σ^2 . More generally, the r th sample moment is given by

$$(1/T) \cdot (x_1^r + x_2^r + \cdots + x_T^r),$$

where x_t^r denotes x_t raised to the r th power.

Bias and Efficiency

Let $\hat{\theta}$ be a sample estimate of a vector of population parameters θ . For example, $\hat{\theta}$ could be the sample mean $\bar{x}$ and θ the population mean μ . The estimate is said to be *unbiased* if $E(\hat{\theta}) = \theta$.

Suppose that $\hat{\theta}$ is an unbiased estimate of θ . The estimate $\hat{\theta}$ is said to be *efficient* if it is the case that for any other unbiased estimate $\hat{\theta}^*$, the following matrix is positive semidefinite:

$$\mathbf{P} \equiv E[(\hat{\theta}^* - \theta) \cdot (\hat{\theta}^* - \theta)'] - E[(\hat{\theta} - \theta) \cdot (\hat{\theta} - \theta)'].$$

Joint Distributions

For two random variables X and Y with the joint density $f_{X,Y}(x, y)$, we calculate the probability of the joint event that both $X \leq a$ and $Y \leq b$ from

$$P\{X \leq a, Y \leq b\} = \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(x, y) dy dx.$$

This can be represented in terms of the joint cumulative distribution function:

$$F_{X,Y}(a, b) = P\{X \leq a, Y \leq b\}.$$

The probability that $X \leq a$ by itself can be calculated from

$$P\{X \leq a, Y \text{ any}\} = \int_{-\infty}^a \left[\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right] dx. \quad [\text{A.5.5}]$$

Comparison of [A.5.5] with [A.5.2] reveals that the *marginal* density $f_X(x)$ is obtained by integrating the joint density $f_{X,Y}(x, y)$ with respect to y :

$$f_X(x) = \left[\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right]. \quad [\text{A.5.6}]$$

Conditional Distributions

The *conditional density* of Y given X is given by

$$f_{Y|X}(y|x) \equiv \begin{cases} \frac{f_{X,Y}(x, y)}{f_X(x)} & \text{if } f_X(x) > 0 \\ 0 & \text{otherwise.} \end{cases} \quad [\text{A.5.7}]$$

Notice that this satisfies the requirement of a density [A.5.1]:

$$\begin{aligned}\int_{-\infty}^{\infty} f_{Y|X}(y|x) dy &= \int_{-\infty}^{\infty} \frac{f_{X,Y}(x, y)}{f_X(x)} dy \\ &= \frac{1}{f_X(x)} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\ &= \frac{f_X(x)}{f_X(x)} = 1.\end{aligned}$$

A further obvious implication of the definition in [A.5.7] is that a joint density can be written as the product of the marginal density and the conditional density:

$$f_{X,Y}(x, y) = f_{Y|X}(y|x) \cdot f_X(x). \quad [\text{A.5.8}]$$

The *conditional expectation* of Y given that the random variable X takes on the particular value x is

$$E(Y|X = x) = \int_{-\infty}^{\infty} y \cdot f_{Y|X}(y|x) dy. \quad [\text{A.5.9}]$$

Law of Iterated Expectations

Note that the conditional expectation is a function of the value of the random variable X . For different realizations of X , the conditional expectation will be a different number. Suppose we view $E(Y|X)$ as a random variable and take its expectation with respect to the distribution of X :

$$E_X[E_{Y|X}(Y|X)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} y \cdot f_{Y|X}(y|x) dy \right] f_X(x) dx.$$

Results [A.5.8] and [A.5.6] can be used to express this expectation as

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \cdot f_{Y,X}(y, x) dy dx = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy.$$

Thus,

$$E_X[E_{Y|X}(Y|X)] = E_Y(Y). \quad [\text{A.5.10}]$$

In words, the random variable $E(Y|X)$ has the same expectation as the random variable Y . This is known as the *law of iterated expectations*.

Independence

The variables Y and X are said to be *independent* if

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y). \quad [\text{A.5.11}]$$

Comparing [A.5.11] with [A.5.8], if Y and X are independent, then

$$f_{Y|X}(y|x) = f_Y(y). \quad [\text{A.5.12}]$$

Covariance

Let μ_X denote $E(X)$ and μ_Y denote $E(Y)$. The *population covariance* between X and Y is given by

$$\text{Cov}(X, Y) \equiv \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) \cdot f_{X,Y}(x, y) dy dx. \quad [\text{A.5.13}]$$

Correlation

The population correlation between X and Y is given by

$$\text{Corr}(X, Y) \equiv \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}.$$

If the covariance (or correlation) between X and Y is zero, then X and Y are said to be *uncorrelated*.

Relation Between Correlation and Independence

Note that if X and Y are independent, then they are uncorrelated:

$$\begin{aligned}\text{Cov}(X, Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) \cdot f_X(x) \cdot f_Y(y) \, dy \, dx \\ &= \int_{-\infty}^{\infty} (x - \mu_X) \left[\int_{-\infty}^{\infty} (y - \mu_Y) \cdot f_Y(y) \, dy \right] \cdot f_X(x) \, dx.\end{aligned}$$

Furthermore,

$$\begin{aligned}\left[\int_{-\infty}^{\infty} (y - \mu_Y) \cdot f_Y(y) \, dy \right] &= \int_{-\infty}^{\infty} y \cdot f_Y(y) \, dy - \mu_Y \cdot \int_{-\infty}^{\infty} f_Y(y) \, dy \\ &= \mu_Y - \mu_Y \\ &= 0.\end{aligned}$$

Thus, if X and Y are independent, then $\text{Cov}(X, Y) = 0$, as claimed.

The converse proposition, however, is not true—the fact that X and Y are uncorrelated is not enough to deduce that they are independent. To construct a counterexample, suppose that Z and Y are independent random variables each with mean zero, and let $X \equiv Z \cdot Y$. Then

$$\begin{aligned}E(X - \mu_X)(Y - \mu_Y) &= E[(ZY) \cdot Y] \\ &= E(Z) \cdot E(Y^2) = 0,\end{aligned}$$

and so X and Y are uncorrelated. They are not, however, independent—the value of ZY depends on Y .

Orthogonality

Consider a sample of size T on two random variables, $\{x_1, x_2, \dots, x_T\}$ and $\{y_1, y_2, \dots, y_T\}$. The two variables are said to be *orthogonal* if

$$\sum_{t=1}^T x_t y_t = 0.$$

Thus, orthogonality is the sample analog of absence of correlation.

For example, let $x_t = 1$ denote a sequence of constants and let $y_t = w_t - \bar{w}$, where $\bar{w} \equiv (1/T) \sum_{t=1}^T w_t$ is the sample mean of the variable w . Then x and y are orthogonal:

$$\sum_{t=1}^T 1 \cdot (w_t - \bar{w}) = \sum_{t=1}^T w_t - T \cdot \bar{w} = 0.$$

Population Moments of Sums

Consider the random variable $aX + bY$. Its mean is given by

$$\begin{aligned} E(aX + bY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (ax + by) \cdot f_{X,Y}(x, y) \, dy \, dx \\ &= a \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \cdot f_{X,Y}(x, y) \, dy \, dx + b \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \cdot f_{X,Y}(x, y) \, dy \, dx \\ &= a \int_{-\infty}^{\infty} x \cdot f_X(x) \, dx + b \int_{-\infty}^{\infty} y \cdot f_Y(y) \, dy, \end{aligned}$$

and so

$$E(aX + bY) = a \cdot E(X) + b \cdot E(Y). \quad [\text{A.5.14}]$$

The variance of $(aX + bY)$ is

$$\begin{aligned} \text{Var}(aX + bY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [(ax + by) - (a\mu_X + b\mu_Y)]^2 \cdot f_{X,Y}(x, y) \, dy \, dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [(ax - a\mu_X)^2 + 2(ax - a\mu_X)(by - b\mu_Y) \\ &\quad + (by - b\mu_Y)^2] \cdot f_{X,Y}(x, y) \, dy \, dx \\ &= a^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)^2 \cdot f_{X,Y}(x, y) \, dy \, dx \\ &\quad + 2ab \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) \cdot f_{X,Y}(x, y) \, dy \, dx \\ &\quad + b^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y - \mu_Y)^2 \cdot f_{X,Y}(x, y) \, dy \, dx. \end{aligned}$$

Thus,

$$\text{Var}(aX + bY) = a^2 \cdot \text{Var}(X) + 2ab \cdot \text{Cov}(X, Y) + b^2 \cdot \text{Var}(Y). \quad [\text{A.5.15}]$$

When X and Y are uncorrelated,

$$\text{Var}(aX + bY) = a^2 \cdot \text{Var}(X) + b^2 \cdot \text{Var}(Y).$$

It is straightforward to generalize results [A.5.14] and [A.5.15]. If $\{X_1, X_2, \dots, X_n\}$ denotes a collection of n random variables, then

$$\begin{aligned} E(a_1X_1 + a_2X_2 + \dots + a_nX_n) \\ = a_1 \cdot E(X_1) + a_2 \cdot E(X_2) + \dots + a_n \cdot E(X_n) \end{aligned} \quad [\text{A.5.16}]$$

$$\begin{aligned} \text{Var}(a_1X_1 + a_2X_2 + \dots + a_nX_n) \\ = a_1^2 \cdot \text{Var}(X_1) + a_2^2 \cdot \text{Var}(X_2) + \dots + a_n^2 \cdot \text{Var}(X_n) \\ + 2a_1a_2 \cdot \text{Cov}(X_1, X_2) + 2a_1a_3 \cdot \text{Cov}(X_1, X_3) + \dots \\ + 2a_1a_n \cdot \text{Cov}(X_1, X_n) + 2a_2a_3 \cdot \text{Cov}(X_2, X_3) + 2a_2a_4 \cdot \text{Cov}(X_2, X_4) \\ + \dots + 2a_{n-1}a_n \cdot \text{Cov}(X_{n-1}, X_n). \end{aligned} \quad [\text{A.5.17}]$$

If the X 's are uncorrelated, then [A.5.17] simplifies to

$$\begin{aligned} \text{Var}(a_1X_1 + a_2X_2 + \cdots + a_nX_n) \\ = a_1^2 \cdot \text{Var}(X_1) + a_2^2 \cdot \text{Var}(X_2) + \cdots + a_n^2 \cdot \text{Var}(X_n). \end{aligned} \quad [\text{A.5.18}]$$

Cauchy-Schwarz Inequality

The *Cauchy-Schwarz inequality* states that for any random variables X and Y whose variances and covariance exist, the correlation is no greater than unity in absolute value:

$$-1 \leq \text{Corr}(X, Y) \leq 1. \quad [\text{A.5.19}]$$

To establish the far right inequality in [A.5.19], consider the random variable

$$Z \equiv \frac{X - \mu_X}{\sqrt{\text{Var}(X)}} - \frac{Y - \mu_Y}{\sqrt{\text{Var}(Y)}}.$$

The square of this variable cannot take on negative values, so

$$E \left[\frac{(X - \mu_X)}{\sqrt{\text{Var}(X)}} - \frac{(Y - \mu_Y)}{\sqrt{\text{Var}(Y)}} \right]^2 \geq 0.$$

Recognizing that $\text{Var}(X)$ and $\text{Var}(Y)$ denote population moments (as opposed to random variables), equation [A.5.15] can be used to deduce

$$\frac{E(X - \mu_X)^2}{\text{Var}(X)} - 2 \left[\frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}} \right] + \frac{E(Y - \mu_Y)^2}{\text{Var}(Y)} \geq 0.$$

Thus,

$$1 - 2 \cdot \text{Corr}(X, Y) + 1 \geq 0,$$

meaning that

$$\text{Corr}(X, Y) \leq 1.$$

To establish the far left inequality in [A.5.19], notice that

$$E \left[\frac{(X - \mu_X)}{\sqrt{\text{Var}(X)}} + \frac{(Y - \mu_Y)}{\sqrt{\text{Var}(Y)}} \right]^2 \geq 0,$$

implying that

$$1 + 2 \cdot \text{Corr}(X, Y) + 1 \geq 0,$$

so that

$$\text{Corr}(X, Y) \geq -1.$$

The Normal Distribution

The variable Y_t has a *Gaussian*, or *Normal*, distribution with mean μ and variance σ^2 if

$$f_{Y_t}(y_t) = \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp \left[\frac{-(y_t - \mu)^2}{2\sigma^2} \right]. \quad [\text{A.5.20}]$$

We write

$$Y_t \sim N(\mu, \sigma^2)$$

to indicate that the density of Y_t is given by [A.5.20].

Centered odd-ordered population moments for a Gaussian variable are zero:

$$E(Y_t - \mu)^r = 0 \quad \text{for } r = 1, 3, 5, \dots$$

The centered fourth moment is

$$E(Y_t - \mu)^4 = 3\sigma^4.$$

Skew and Kurtosis

The skewness of a variable Y_t with mean μ is represented by

$$\frac{E(Y_t - \mu)^3}{[\text{Var}(Y_t)]^{3/2}}.$$

A variable with a negative skew is more likely to be far below the mean than it is to be far above the mean. The *kurtosis* is

$$\frac{E(Y_t - \mu)^4}{[\text{Var}(Y_t)]^2}.$$

A distribution whose kurtosis exceeds 3 has more mass in the tails than a Gaussian distribution with the same variance.

Other Useful Univariate Distributions

Let $(X_1, X_2, \dots, X_n)$ be independent and identically distributed (i.i.d.) $N(0, 1)$ variables, and consider the sum of their squares:

$$Y = X_1^2 + X_2^2 + \dots + X_n^2.$$

Then Y is said to have a *chi-square* distribution with n degrees of freedom, denoted

$$Y \sim \chi^2(n).$$

Let $X \sim N(0, 1)$ and $Y \sim \chi^2(n)$ with X and Y independent. Then

$$Z = \frac{X}{\sqrt{Y/n}}$$

is said to have a *t distribution* with n degrees of freedom, denoted

$$Z \sim t(n).$$

Let $Y_1 \sim \chi^2(n_1)$ and $Y_2 \sim \chi^2(n_2)$ with Y_1 and Y_2 independent. Then

$$Z = \frac{Y_1/n_1}{Y_2/n_2}$$

is said to have an *F distribution* with n_1 numerator degrees of freedom and n_2 denominator degrees of freedom, denoted

$$Z \sim F(n_1, n_2).$$

Note that if $Z \sim t(n)$, then $Z^2 \sim F(1, n)$.

Likelihood Function

Suppose we have observed a sample of size T on some random variable Y_t . Let $f_{Y_1, Y_2, \dots, Y_T}(y_1, y_2, \dots, y_T; \theta)$ denote the joint density of $Y_1, Y_2, \dots, Y_T$.

The notation emphasizes that this joint density is presumed to depend on a vector of population parameters θ . If we view this joint density as a function of θ (given the data on Y), the result is called the *sample likelihood function*.

For example, consider a sample of T i.i.d. variables drawn from a $N(\mu, \sigma^2)$ distribution. For this distribution, $\theta = (\mu, \sigma^2)'$, and from [A.5.11] the joint density is the product of individual terms such as [A.5.20]:

$$f_{Y_1, Y_2, \dots, Y_T}(y_1, y_2, \dots, y_T; \mu, \sigma^2) = \prod_{i=1}^T f_{Y_i}(y_i; \mu, \sigma^2).$$

The log of the joint density is the sum of the logs of these terms:

$$\begin{aligned} \log f_{Y_1, Y_2, \dots, Y_T}(y_1, y_2, \dots, y_T; \mu, \sigma^2) \\ &= \sum_{i=1}^T \log f_{Y_i}(y_i; \mu, \sigma^2) \\ &= (-T/2) \log(2\pi) - (T/2) \log(\sigma^2) - \sum_{i=1}^T \frac{(y_i - \mu)^2}{2\sigma^2}. \end{aligned} \quad [\text{A.5.21}]$$

Thus, for a sample of T Gaussian random variables with mean μ and variance σ^2 , the sample log likelihood function, denoted $\mathcal{L}(\mu, \sigma^2; y_1, y_2, \dots, y_T)$, is given by:

$$\mathcal{L}(\mu, \sigma^2; y_1, y_2, \dots, y_T) = k - (T/2) \log(\sigma^2) - \sum_{i=1}^T \frac{(y_i - \mu)^2}{2\sigma^2}. \quad [\text{A.5.22}]$$

In calculating the sample log likelihood function, any constant term that does not involve the parameter μ or σ^2 can be ignored for most purposes. In [A.5.22], this constant term is

$$k \equiv -(T/2) \log(2\pi).$$

Maximum Likelihood Estimation

For a given sample of observations $(y_1, y_2, \dots, y_T)$, the value of θ that makes the sample likelihood as large as possible is called the *maximum likelihood estimate (MLE)* of θ . For example, the maximum likelihood estimate of the population mean μ for an i.i.d. sample of size T from a $N(\mu, \sigma^2)$ distribution is found by setting the derivative of [A.5.22] with respect to μ equal to zero:

$$\frac{\partial \mathcal{L}}{\partial \mu} = \sum_{i=1}^T \frac{y_i - \mu}{\sigma^2} = 0,$$

or

$$\hat{\mu} = (1/T) \sum_{i=1}^T y_i. \quad [\text{A.5.23}]$$

The MLE of σ^2 is characterized by

$$\frac{\partial \mathcal{L}}{\partial \sigma^2} = -\frac{T}{2\sigma^2} + \sum_{i=1}^T \frac{(y_i - \mu)^2}{2\sigma^4} = 0. \quad [\text{A.5.24}]$$

Substituting [A.5.23] into [A.5.24] and solving for σ^2 gives

$$\hat{\sigma}^2 = (1/T) \sum_{i=1}^T (y_i - \hat{\mu})^2. \quad [\text{A.5.25}]$$

Thus, the sample mean is the *MLE* of the population mean and the sample variance is the *MLE* of the population variance for an i.i.d. sample of Gaussian variables.

Multivariate Gaussian Distribution

Let

$$\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)'$$

be a collection of n random variables. The vector $\mathbf{Y}$ has a *multivariate Normal*, or *multivariate Gaussian*, distribution if its density takes the form

$$f_{\mathbf{Y}}(\mathbf{y}) = (2\pi)^{-n/2} |\mathbf{\Omega}|^{-1/2} \exp\left[(-1/2)(\mathbf{y} - \boldsymbol{\mu})' \mathbf{\Omega}^{-1} (\mathbf{y} - \boldsymbol{\mu})\right]. \quad [\text{A.5.26}]$$

The mean of $\mathbf{Y}$ is given by the vector $\boldsymbol{\mu}$:

$$E(\mathbf{Y}) = \boldsymbol{\mu};$$

and its variance-covariance matrix is $\mathbf{\Omega}$:

$$E(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})' = \mathbf{\Omega}.$$

Note that $(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})'$ is symmetric and positive semidefinite for any $\mathbf{Y}$, meaning that any variance-covariance matrix must be symmetric and positive semidefinite; the form of the likelihood in [A.5.26] assumes that $\mathbf{\Omega}$ is positive definite.

Result [A.4.15] is sometimes used to write the multivariate Gaussian density in an equivalent form:

$$f_{\mathbf{Y}}(\mathbf{y}) = (2\pi)^{-n/2} |\mathbf{\Omega}^{-1}|^{1/2} \exp\left[(-1/2)(\mathbf{y} - \boldsymbol{\mu})' \mathbf{\Omega}^{-1} (\mathbf{y} - \boldsymbol{\mu})\right].$$

If $\mathbf{Y} \sim N(\boldsymbol{\mu}, \mathbf{\Omega})$, then for any nonstochastic $(r \times n)$ matrix $\mathbf{H}'$ and $(r \times 1)$ vector $\mathbf{b}$,

$$\mathbf{H}'\mathbf{Y} + \mathbf{b} \sim N((\mathbf{H}'\boldsymbol{\mu} + \mathbf{b}), \mathbf{H}'\mathbf{\Omega}\mathbf{H}).$$

Correlation and Independence for Multivariate Gaussian Variates

If $\mathbf{Y}$ has a multivariate Gaussian distribution, then absence of correlation implies independence. To see this, note that if the elements of $\mathbf{Y}$ are uncorrelated, then $E[(Y_i - \mu)(Y_j - \mu)] = 0$ for $i \neq j$ and the off-diagonal elements of $\mathbf{\Omega}$ are zero:

$$\mathbf{\Omega} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{bmatrix}.$$

For such a diagonal matrix $\mathbf{\Omega}$,

$$|\mathbf{\Omega}| = \sigma_1^2 \sigma_2^2 \cdots \sigma_n^2 \quad [\text{A.5.27}]$$

$$\mathbf{\Omega}^{-1} = \begin{bmatrix} 1/\sigma_1^2 & 0 & \cdots & 0 \\ 0 & 1/\sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sigma_n^2 \end{bmatrix}. \quad [\text{A.5.28}]$$

Substituting [A.5.27] and [A.5.28] into [A.5.26] produces

$$\begin{aligned} f_{\mathbf{Y}}(\mathbf{y}) &= (2\pi)^{-n/2} [\sigma_1^2 \sigma_2^2 \cdots \sigma_n^2]^{-1/2} \\ &\quad \times \exp\left\{(-1/2)\{(y_1 - \mu_1)^2/\sigma_1^2 + (y_2 - \mu_2)^2/\sigma_2^2 + \cdots \right. \\ &\quad \left. + (y_n - \mu_n)^2/\sigma_n^2\}\right\} \\ &= \prod_{i=1}^n (2\pi)^{-1/2} [\sigma_i^2]^{-1/2} \exp\{(-1/2)\{(y_i - \mu_i)^2/\sigma_i^2\}\}, \end{aligned}$$

which is the product of n univariate Gaussian densities. Since the joint density is the product of the individual densities, the random variables $(Y_1, Y_2, \dots, Y_n)$ are independent.

Probability Limit

Let $\{X_1, X_2, \dots, X_T\}$ denote a sequence of random variables. Often we are interested in what happens to this sequence as T becomes large. For example, X_T might denote the sample mean of T observations:

$$X_T = (1/T) \cdot (Y_1 + Y_2 + \cdots + Y_T), \quad [\text{A.5.29}]$$

in which case we might want to know the properties of the sample mean as the size of the sample T grows large.

The sequence $\{X_1, X_2, \dots, X_T\}$ is said to *converge in probability* to c if for every $\varepsilon > 0$ and $\delta > 0$ there exists a value N such that, for all $T \geq N$,

$$P\{|X_T - c| > \delta\} < \varepsilon. \quad [\text{A.5.30}]$$

When [A.5.30] is satisfied, the number c is called the *probability limit*, or *plim*, of the sequence $\{X_1, X_2, \dots, X_T\}$. This is sometimes indicated as

$$X_T \xrightarrow{p} c.$$

Law of Large Numbers

Under fairly general conditions detailed in Chapter 7, the sample mean [A.5.29] converges in probability to the population mean:

$$(1/T)(Y_1 + Y_2 + \cdots + Y_T) \xrightarrow{p} E(Y_1). \quad [\text{A.5.31}]$$

When [A.5.31] holds, we say that the sample mean gives a *consistent* estimate of the population mean.

Convergence in Mean Square

A stronger condition than convergence in probability is *mean square convergence*. The sequence $\{X_1, X_2, \dots, X_T\}$ is said to converge in mean square if for every $\varepsilon > 0$ there exists a value N such that, for all $T \geq N$,

$$E(X_T - c)^2 < \varepsilon. \quad [\text{A.5.32}]$$

We indicate that the sequence converges to c in mean square as follows:

$$X_T \xrightarrow{m.s.} c.$$

Convergence in mean square implies convergence in probability, but convergence in probability does not imply convergence in mean square.

Appendix A References

- Chiang, Alpha C. 1974. *Fundamental Methods of Mathematical Economics*, 2d ed. New York: McGraw-Hill.
- Hoel, Paul G., Sidney C. Port, and Charles J. Stone. 1971. *Introduction to Probability Theory*. Boston: Houghton Mifflin.
- Johnston, J. 1984. *Econometric Methods*, 3d ed. New York: McGraw-Hill.
- Lindgren, Bernard W. 1976. *Statistical Theory*, 3d ed. New York: Macmillan.
- Magnus, Jan R., and Heinz Neudecker. 1988. *Matrix Differential Calculus with Applications in Statistics and Econometrics*. New York: Wiley.
- Marsden, Jerrold E. 1974. *Elementary Classical Analysis*. San Francisco: Freeman.
- O’Nan, Michael. 1976. *Linear Algebra*, 2d ed. New York: Harcourt Brace Jovanovich.
- Strang, Gilbert. 1976. *Linear Algebra and Its Applications*. New York: Academic Press.
- Theil, Henri. 1971. *Principles of Econometrics*. New York: Wiley.
- Thomas, George B., Jr. 1972. *Calculus and Analytic Geometry*, alternate ed. Reading, Mass.: Addison-Wesley Publishing Company, Inc.